Vulnerabilities of Manual Reservation Systems

In partial fulfillment of the requirements for Capstone 1 and Research Subject

Jayson Sapio, Aldave Sadiang-ibon, Alexander Paller, Patrick Shane Sevigan,

John Edward Tiangson

College of Information Technology Education, PHINMA University of Iloilo, Philippines

Abstract

[1] The present manual processes for resort reservations result in delayed processes, hassled reservations, and the possibility of human error. The researchers presented the RMS as a non-contact and efficient method to overcome these issues. The system was designed and developed using agile methodology, including iterative and incremental development procedures. As a result, the developed system transforms and improves the resort's booking and management processes. This system provides a simple and user-friendly online interface for customers to book rooms and cottages effortlessly. Customers can also view the resort's available amenities and services. The developed system includes real-time availability updates, assuring accurate booking information while reducing the possibility of overbooking. Furthermore, the management feature allows Nature Spring Resort owners to handle reservations and monitor guest preferences easily. 3 According to the IT experts' evaluation results, which were based on the ISO 25010:2015 standards for quality software, the developed system has exceptional functionality, reliability, usability, efficiency, maintainability, portability, and compliance. This suggests that the management of the Nature Spring Resort, especially in the booking and reservation processes, can greatly benefit from the developed system. With these findings, the implementation of this developed system has come up with a solution to improve Nature Spring Resort's operational efficiency, customer satisfaction, and overall management effectiveness. Thus, we strongly recommend that this system facilitate the resort's business processing.

Index Keywords - Web-Based Reservation System, CoravergelResort, Booking System, Management System

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Digital technologies are now a standard part of how hotels and resorts operate. A systematic review of 79 studies found that digital technologies contribute to hotel performance in six areas: financial results, competitiveness, quality of service, resource utilization, flexibility, and innovation [1]. Digital transformation, which is the integration of digital tools into everyday business operations, has also been documented as a major driver of change in hotel management practices worldwide [2]. Among these tools, reservation systems are especially important because they sit at the center of guest interaction, room allocation, and record keeping.

Despite this trend, many small and medium-sized lodging businesses still manage reservations manually. Recent studies describe the same pattern in different settings. At Hotel Ariesta in Indonesia, reservations were recorded in a logbook, and room availability information was not integrated, which exposed the hotel to recording errors, data loss, and double bookings. After a web-based reservation system replaced the manual process, the researchers reported shorter processing time and fewer recording errors [3]. A similar situation was reported at a beach hotel in Takalar, Indonesia, where room availability could only be obtained through the receptionist, transaction fees were computed by hand, and memoranda were written on paper. A web-based room reservation system gave administrators accurate and timely information on room availability [4]. Another study implemented a web-based reservation system for a hotel using PHP and MySQL, allowing the receptionist to view all reservations and the administrator to view, delete, and assign them [5].

Coravergel Inland Resort is one of the resorts that provides relaxation, recreation, and accommodation services to customers. The resort receives guests who want to enjoy leisure activities, celebrate events, or spend quality time with family and friends. As the number of customers increases, the management of reservations becomes more important and challenging. At present, Coravergel Inland Resort still uses a manual reservation process to manage bookings and customer records. Reservations are commonly recorded using paper-based documents, notebooks, spreadsheets, or basic computer files. Although this method may still work for a small number of reservations, it becomes difficult and inefficient when the volume of customers increases.

The use of a manual reservation system creates several problems for the resort. One major issue is the possibility of human error. Since information is manually written or encoded, staff may accidentally record incorrect details, forget reservations, or duplicate bookings. These mistakes may result in conflicts between customers and management, causing inconvenience and dissatisfaction among guests. Overbooking may occur when two or more customers are assigned to the same room or schedule due to inaccurate monitoring of reservations. In addition, manual systems make it difficult to update room availability in real time, which can confuse staff when accommodating walk-in guests or advance reservations.

Another problem with manual reservation systems is inefficiency. Staff members spend a significant amount of time searching for reservation records, updating information, and confirming bookings. This slows down operations and increases workload, especially during busy seasons or holidays when customer demand is high. Delays in processing reservations can affect the quality of customer service because guests expect quick responses and accurate information regarding their bookings. A slow reservation process may also reduce customer trust and satisfaction, which can negatively affect the reputation of the resort.

Security is also a major concern in manual reservation systems. Customer information such as names, contact details, schedules, and reservation records are important and confidential data that must be protected. When records are stored manually or in unsecured files, there is a higher risk of data loss, unauthorized access, and damage to information. Paper records can easily be misplaced, destroyed, or accessed by unauthorized individuals. In some cases, there is no proper monitoring of who accesses or changes reservation information. Without security measures such as authentication, access control, and activity monitoring, the resort becomes vulnerable to data breaches and information misuse.

The growing dependence on digital technology in the hospitality industry highlights the need for a more advanced and secure reservation management system. Modern web-based reservation systems provide several advantages compared to manual processes. These systems can automate booking transactions, update room availability instantly, organize records efficiently, and improve communication between staff and customers. In addition, digital systems can include security features such as user authentication, password protection, role-based access control, and encrypted data storage to ensure that customer information remains safe and protected.

Studies conducted in both foreign and local settings have shown that computerized reservation systems significantly improve operational efficiency and customer satisfaction. Research also indicates that automated systems reduce booking errors, save time, and improve data management. However, some existing systems focus mainly on automation and convenience while giving less attention to cybersecurity and data protection. This creates a need for a reservation system that not only improves efficiency but also prioritizes the security and confidentiality

of customer information. Because of these issues, the researchers decided to conduct this study to analyze the vulnerabilities of the current manual reservation system used at Coravergel Inland Resort and develop a secure web-based reservation management system. The proposed system aims to improve reservation processing, provide real-time room monitoring, minimize booking conflicts, and strengthen the protection of customer data. The researchers also aim to create a system that is easy to use for resort staff and administrators while ensuring accuracy, reliability, and efficiency in daily operations.

Coravergel Inland Resort provides relaxation, recreation, and accommodation services to guests who visit for leisure, family gatherings, and special events. At present, the resort still manages bookings and guest records through paper documents, notebooks, spreadsheets, and basic computer files. This approach may be workable for a small number of bookings, but it becomes inefficient and error-prone as the number of guests grows, especially during peak seasons and holidays. Manual encoding can lead to duplicate bookings, incorrect guest details, and missing records. Because room status is not updated in real time, staff may also promise the same room to more than one guest.

Manual reservation handling also creates information security problems. Reservation records contain personal data such as names, contact details, schedules, and accommodation preferences. Hospitality businesses hold large volumes of this kind of sensitive data, which makes them attractive targets for cybercriminals. Industry analysis of 2021 to 2022 data reported an average data breach cost of USD 2.94 million in the hospitality sector, and a September 2022 attack on a multinational hospitality company caused a two-day outage of its online booking system [6]. Interviews with hotel managers in a recent study led to the conclusion that hotels must prioritize cybersecurity by carrying out risk assessments and applying technical and organizational measures that mitigate threats and protect information [7]. Paper records and unsecured files cannot provide these controls. They have no user authentication, no access restrictions by role, no activity tracking, and no reliable backup.

In the Philippines, the protection of personal data is governed by the Data Privacy Act of 2012 (Republic Act No. 10173), which is enforced by the National Privacy Commission (NPC). A 2023 review of NPC circulars, advisories, and advisory opinions showed that technology adoption accelerated during the COVID-19 pandemic and recommended embedding privacy into the design lifecycle of systems, with end-to-end security as one of the guiding principles [8]. For a resort that collects guest information, this means that a reservation system should be designed with security and privacy in mind from the start.

The studies above show that web-based reservation systems improve speed, accuracy, and accessibility [3]–[5]. However, most of them focus on automation and convenience, while cybersecurity concerns, such as authentication, access control, and audit trails, receive less attention, even though hospitality businesses continue to face serious

cyber threats [6], [7]. This gap motivates the present study. The researchers aim to analyze the vulnerabilities of the manual reservation system used at Coravergel Inland Resort and to develop a secure, centralized, web-based reservation management system that reduces booking errors, provides real-time room monitoring, and protects guest data.

1.2 Statement of the Problem

1.2.1 General Problem

Coravergel Inland Resort relies on a manual reservation process to manage bookings, room assignments, and guest records. This process is slow, prone to human error, and weak in terms of security and accountability. Coravergel Inland Resort currently relies on a manual reservation system for managing customer bookings, room assignments, and guest records. While this traditional method has been used for some time, it presents several operational challenges that affect the efficiency, accuracy, security, and overall effectiveness of the resort's daily activities. As the hospitality industry continues to embrace digital transformation, relying on manual processes becomes increasingly difficult and less practical in meeting the growing demands of customers and business operations.

One of the primary issues is inefficiency in handling reservations. Since the system depends on manual encoding and paper-based documentation, staff members spend a significant amount of time recording, updating, and searching for reservation details. This slows down the booking process and increases workload, especially during peak seasons when the number of customers is high. Instead of focusing on customer service and operational tasks, employees are often occupied with administrative work related to reservation management. The manual process also increases waiting times for customers who seek immediate confirmation or information regarding room availability.

Another major concern is the high risk of human error. Manual entry of reservation details often leads to mistakes such as duplicate bookings, incorrect guest information, missing records, and inaccurate room assignments. Human errors may occur due to fatigue, lack of attention, miscommunication, or improper record management. These errors can create confusion among staff members and negatively affect customer experiences. In some cases, customers may arrive at the resort only to discover that their reservations were incorrectly recorded or that their assigned rooms are unavailable. Such situations can damage customer trust and negatively impact the reputation of the resort.

The current system also lacks real-time monitoring of room availability. Because updates are not automatically synchronized, staff may not always have accurate information regarding which rooms are available, occupied, reserved, or under maintenance. This limitation creates delays in responding to customer inquiries and increases the possibility of overbooking. Without real-time updates, multiple staff members may unknowingly process reservations for the same room, resulting in scheduling conflicts and operational disruptions. Real-time monitoring is essential in ensuring that reservation information remains accurate and consistent at all times.

.As guest expectations rise and the hospitality industry continues to digitize [1], [2], the resort needs a secure and centralized system that can replace its manual process.

1.2.2 Specific Problems

1. The manual reservation process requires heavy paperwork and manual encoding, which makes booking slow and time-consuming for staff and management.
2. Reservation records are scattered across paper documents, notebooks, and spreadsheets, so searching, updating, and verifying information takes too much time and leads to inconsistent or duplicate data.
3. Manual encoding increases the risk of human error, such as duplicate bookings, incorrect guest information, incomplete records, and scheduling conflicts.
4. The resort has no real-time room availability monitoring, so staff cannot immediately tell which rooms are available, reserved, occupied, or under maintenance. This results in overbooking and confusion when handling inquiries and walk-in guests.
5. Guest information is not securely stored, which exposes it to loss, damage, unauthorized access, and accidental deletion.
6. The current process has no user authentication, password protection, or role-based access control to restrict who can view or change reservation records.
7. There are no activity logs or audit trails, so management cannot identify who created, modified, or deleted a reservation, and accountability is weak.
8. There is no reliable backup and recovery mechanism, so reservation data cannot be restored if documents are lost, damaged, or corrupted.
9. Report generation for bookings, occupancy, and reservation activity is manual, which delays management decisions during planning and peak seasons.
10. Staff communication about reservations and room status is difficult because there is no centralized platform where information can be updated and viewed instantly.

1.3 CONCEPTUAL FRAMEWORK

The study uses an IPO model with feedback. The input is the problem situation and the resources, the process is the development and testing of the system, and the output is the developed system and its expected effects. Evaluation results return to the process so that the system can be improved.

TABLE I. INPUT-PROCESS-OUTPUT OF THE STUDY

Input	Process	Output
Vulnerabilities of the manual process: duplicate bookings, data entry errors, no real-time availability, scattered records, no authentication or access control, no audit trail, no backup, slow reporting	Requirements gathering through interviews, surveys, observation, and document analysis	Secure web-based reservation management system
User requirements from administrators, reservation staff, and front desk personnel	System design: three-tier architecture, ERD, DFD, use case diagrams, security framework	Centralized database of reservations, guests, rooms, users, and logs
Technical resources: PHP, MySQL, Apache (XAMPP), HTML, CSS, JavaScript, Bootstrap, Visual Studio Code, standard desktop or laptop computers	Development of modules: authentication, dashboard, customer, reservation, room, user, reports, activity log, and backup	Accurate reservations and real-time room availability
Security principles: CIA triad, RBAC, least privilege, OWASP risks, NIST guidance [21]-[24]	Testing: functional, integration, performance, security, and user acceptance testing	Controlled access, hashed credentials, and activity logs for accountability

Standards and related studies [9]-[16]	Deployment, staff orientation, and maintenance	Automated reports and recoverable records

Feedback loop: results of testing, user acceptance, and evaluation are used to correct errors and refine the modules before and after deployment.

1.3.1 Input

The input has four parts. The first is the set of vulnerabilities found in the manual reservation process, which define the problem. The second is the set of user requirements gathered from resort personnel. The third is the technical resources used to build the system. The fourth is the security principles and standards that guide the design [3]-[6].

1.3.2 Process

The process covers requirements gathering, design, construction, testing, deployment, and maintenance. Security is built into each phase instead of added at the end. During design, the roles Administrator and Reservation Staff are defined and a permission set is assigned to each. During construction, input validation, duplicate-reservation checking, password hashing, session timeout, and activity logging are implemented. During testing, authentication, access control, and logging are verified, including attempts by test accounts to open restricted functions.

1.3.3 Output

The output is a secure web-based reservation management system with a centralized database. Its expected effects are fewer booking errors, immediate room availability information, protected guest records, traceable user actions, automated reports, and recoverable data.

1.4 Mapping of Vulnerabilities, Controls, and Evaluation

Table II links each weakness of the manual process to the control that addresses it, the CIA property it supports, and the evidence used to evaluate it.

TABLE II. MAPPING OF VULNERABILITIES TO CONTROLS AND EVALUATION

Manual-process vulnerability	Control in the system	CIA property	Evaluation evidence
Paper and file records can be read or taken by unauthorized persons	Login authentication, hashed passwords, RBAC	Confidentiality	Access-control test cases for each role
Duplicate bookings and wrong or incomplete entries	Input validation, availability check, database constraints	Integrity	Duplicate-booking and validation tests
No record of who created, changed, or deleted a reservation	Activity logs with user, action, date, time, and IP address	Integrity and accountability	Log review during testing
Records lost, damaged, or misplaced	Centralized database, backup and recovery	Availability	Backup and restore test
Outdated room availability	Real-time room status updates	Integrity and availability	Synchronization and response-time tests
Slow manual report preparation	Automated reservation, occupancy, and activity reports	Efficiency	Report generation time and user feedback

This mapping follows the same logic as the OWASP and NIST references: access control and authentication reduce unauthorized access [11]-[13], logging supports detection of misuse [9], [12], [14], and backup supports recovery [14].

1.5 RELATIONSHIP OF VARIABLES

The independent variable is the secure web-based reservation management system, represented by its features: centralized data, real-time availability, automation, access control, and activity logging. The dependent variables are reservation accuracy, processing efficiency, security of guest information, accountability, and user satisfaction. The study assumes that using the system improves these variables compared with the manual process. They are measured through functionality, usability, reliability, efficiency, and security ratings, consistent with the quality characteristics in [15].

1.6SCOPE OF THE FRAMEWORK

The framework covers reservation management at Coravergel Inland Resort for authorized resort personnel. Online payment, mobile applications, third-party booking platforms, multi-factor authentication, and advanced security tools are outside the scope of the study and are treated as future enhancements. Recent work such as [9] already uses two-factor authentication, which shows a direction for future versions.

1.7 Objectives of the Study

1.7.1 General Objective

The general objective of this study is to design, develop, and evaluate a secure, centralized, web-based reservation management system for Coravergel Inland Resort that replaces the manual reservation process and protects the confidentiality, integrity, and availability of guest information.

This study aims to address the operational limitations and vulnerabilities associated with the current manual reservation process. The proposed system intends to minimize the dependence on paper-based records and manual encoding by automating reservation-related transactions and record management activities. By reducing human intervention in routine reservation tasks, the system is expected to decrease the occurrence of common errors such as duplicate bookings, incorrect customer information, incomplete records, and scheduling conflicts. This will contribute to a more accurate, consistent, and dependable reservation management process.

Furthermore, the study seeks to establish a centralized database that will serve as the primary repository of all reservation and guest information. Through centralized data management, staff members and administrators will be able to access, update, retrieve, and monitor reservation records more efficiently. The centralized approach will eliminate data fragmentation, improve record organization, and ensure that information remains accurate and readily available whenever needed. This feature is expected to significantly reduce the time spent searching for records and improve overall administrative productivity.

Another important objective of the study is to enhance operational efficiency through automation and real-time information processing. The proposed system will provide real-time monitoring of room availability, allowing authorized users to instantly determine the status of rooms as available, occupied, reserved, or unavailable. This functionality will help prevent overbooking, reduce reservation conflicts, and improve the speed and accuracy of reservation transactions. By providing immediate access to updated information, the system can support faster decision-making and more effective resource allocation.

The study also aims to strengthen the cybersecurity posture of the resort by incorporating security mechanisms that protect sensitive customer information and reservation records. The proposed system will implement user authentication, password protection, role-based access control, activity logging, and secure data storage practices to safeguard information from unauthorized access, modification, disclosure, or loss. These security measures are intended to support the principles of confidentiality, integrity, and availability of information, which are essential components of cybersecurity and information assurance.

1.7.2 Specific Objectives

1. To develop a user-friendly web-based system that manages reservations, room availability, and guest information in one centralized database.
2. To automate the booking process in order to reduce manual work and speed up reservation transactions.
3. To provide real-time monitoring of room status (available, reserved, occupied, or unavailable) to prevent overbooking and scheduling conflicts.
4. To reduce duplicate bookings, incomplete records, and inconsistent data through automated validation.
5. To implement user authentication, password protection, and role-based access control so that only authorized personnel can access the system.
6. To protect guest and reservation records through secure data storage and controlled system access.
7. To record user actions through activity logs and audit trails that improve accountability and transparency.
8. To provide backup and recovery functions that help prevent data loss and support business continuity.
9. To generate accurate reservation, occupancy, and transaction reports that support management decisions.

10. To test the system for functionality, usability, reliability, and security, and to follow established software development practices and data privacy principles [8].

1.8 Scope and Limitations

1.8.1 Scope of the Study

This study focuses on the development of a secure, efficient, and centralized web-based reservation system for Coravergel Inland Resort. The system is designed to modernize and improve the resort's current manual reservation process by introducing a digital platform that handles booking operations in a more organized, accurate, and time-efficient manner

The scope of the system includes key functionalities that support the core operations of reservation management. First, the system allows users to create, view, update, and manage room reservations through a centralized interface. This ensures that all booking information is stored in a single database, reducing redundancy and improving data organization. Second, the system provides real-time room availability tracking, enabling staff to immediately determine whether a room is available, reserved, occupied, or temporarily unavailable. This feature helps prevent overbooking, scheduling conflicts, and reservation errors that commonly occur in manual reservation processes.

The system includes features for managing guest information, ensuring that customer data is properly recorded, stored, updated, and retrieved whenever necessary. The guest management module is designed to maintain accurate customer records while reducing the risks associated with manual data handling. The system also incorporates basic security mechanisms such as user authentication, login validation, password protection, and role-based access control to ensure that only authorized users can access, modify, or manage reservation information.

The proposed system includes activity logging and audit trail features that record user actions performed within the system. These monitoring capabilities allow administrators to track reservation-related activities, identify unauthorized actions, and improve accountability among users. The logging mechanism also provides management with a transparent view of system usage and supports operational monitoring and security management.

The study also covers the development of a web-based interface that is designed to be simple, user-friendly, responsive, and accessible to resort staff and administrators. The interface is intended to improve usability by providing an organized layout, intuitive navigation, and efficient access to reservation functions. Through a web-based environment, authorized users can access the system using computers connected to the resort's local network or internet connection, depending on deployment requirements.

The proposed system is designed to automate several reservation-related processes, including reservation creation, reservation confirmation, room assignment management, record retrieval, and reservation status monitoring. By

automating these tasks, the system aims to reduce the amount of manual work required by employees while improving processing speed and operational efficiency. Automation also contributes to reducing human errors that may occur during data entry, record management, and reservation verification.

1. Reservation management. This is the core module. Staff can create a reservation by selecting a guest, a room, check-in and check-out dates, and the number of guests. The system checks the room's availability for those dates and rejects any booking that overlaps an existing one, which prevents double booking. Staff can view all reservations in one list, update details such as dates or special requests, confirm a pending booking, or cancel it. Each reservation carries a status that shows where it is in its life cycle:

- *Pending:* submitted but not yet approved.
- *Confirmed:* approved, and the room is held for the guest.
- *Checked-in:* the guest has arrived and occupies the room.
- *Checked-out:* the guest has left and the stay is closed.
- *Completed:* the stay is finished and its records are final.
- *Cancelled:* the booking was withdrawn, and the room becomes available again.

Tracking status this way gives staff an accurate picture of current operations and a clear history of every booking.

2. Room management. The system stores each room's number, category (for example standard, family, or cottage), maximum capacity, rate, description, and current status (available, reserved, occupied, or under maintenance). When a reservation is created, changed, or cancelled, the room's status updates automatically, so the availability display always matches the bookings. Staff can see at a glance which rooms are free for a chosen date range, which prevents assigning a room to two guests or offering a room that is under maintenance.

3. Guest management. Every guest has a profile with contact details, address, and an ID reference, stored once in a centralized database. The profile is linked to all of that guest's reservations, so staff can see past stays, preferences, and special requests. Returning guests need not be re-entered, which saves time and prevents duplicate or inconsistent records. Records can be updated and retrieved quickly, replacing the paper forms and logbooks used before.

4. User management and security. Administrators create and manage staff accounts and assign each a role. The security controls work together as follows:

- *Secure login:* every user signs in with a username and password before reaching any function.

- *Password protection:* passwords are stored as salted hashes, not plain text, so a database leak does not expose them.
- *Session management:* a signed-in user gets a session that expires after a period of inactivity, which limits misuse of unattended computers.
- *Role-based access control (RBAC) with least privilege:* each role sees and uses only the functions its duties need. For example, front-desk staff manage reservations but cannot create user accounts or change system settings.

These controls support the CIA principles. Confidentiality comes from authentication and access control, integrity from restricted editing and validation, and availability from reliable access and backups.

5. Activity logging and audit trails. The system records each significant action with the user, the action, the date and time, and the IP address where available. The logged actions are logins, reservation creation, updates and cancellations, and administrative actions such as account changes. This gives administrators an audit trail for investigating errors or suspicious activity, and it makes staff accountable for their actions. A log entry should not be altered by ordinary users, since a log that can be edited loses its value as evidence.

6. Search, filtering, and reporting. Staff can search and filter by guest name, date, room number, status, or booking reference, so a record can be found in seconds instead of by paging through files. Administrators can generate three kinds of report:

- *Reservation summaries:* bookings for a chosen period, grouped by status.
- *Occupancy reports:* how many rooms were occupied or available over time.
- *Activity reports:* the logged user actions.

These reports replace manual compilation and give management organized, current data for decisions.

7. Backup and recovery. The system supports periodic backup of reservation records and related data to external storage or a secure cloud location. If data is lost through hardware failure, accidental deletion, or a software fault, the latest backup can be restored, so operations resume with limited loss. Regular backups support the availability principle, and backup files should be stored apart from the main server so that one failure does not destroy both.

8. Testing and evaluation. The developed system is tested in four ways:

- *Functional testing:* confirms that each feature, such as login, booking, and reports, gives the expected output.

- *Usability testing:* checks that staff can learn and use the interface easily.
- *Reliability testing:* checks that the system works consistently under normal use.
- *Security testing:* confirms that authentication, access control, and session handling block unauthorized access, using test accounts.

1.8.2 Limitations of the Study

This study is limited to the development of a web-based reservation management system and does not extend to other advanced or external business functionalities. The system is strictly focused on internal reservation processing and does not include features that fall outside the primary objective of managing bookings, room availability, customer information, and reservation records.

The system does not support online payment processing or integration with electronic payment gateways such as GCash, PayPal, Maya, credit card processing systems, online banking services, or financial APIs. All payment-related activities, including payment verification, receipt generation, refund processing, and financial transaction monitoring, are excluded from the scope of this study. Any payment transactions associated with reservations will continue to follow the existing procedures established by the resort.

The system is not designed as a mobile application. The study focuses exclusively on the development of a web-based platform that can be accessed through standard web browsers using desktop computers, laptops, or compatible devices. Although the user interface may be optimized for different screen sizes to improve accessibility, the development of dedicated Android, iOS, or cross-platform mobile applications is beyond the scope of this research.

The system also does not integrate with external booking platforms such as Agoda, Booking.com, Airbnb, Expedia, Trivago, Traveloka, or other third-party reservation services. Reservation information will be managed solely within the internal system of Coravergel Inland Resort. Synchronization of reservations from external travel agencies, online travel management systems, or external booking channels is not included in the implementation of the proposed system.

the system is limited to basic reporting and monitoring functions. While administrators can generate reservation summaries, occupancy reports, and reservation activity reports, the system does not include advanced business intelligence tools, predictive analytics, machine learning algorithms, artificial intelligence-based forecasting, customer behavior analysis, or data mining capabilities. The primary objective is to improve reservation management rather than perform complex business analysis.

The study does not include the development of an inventory management system for resort resources. Monitoring and management of supplies, equipment, maintenance materials, food inventories, beverages, and other operational assets are not part of the proposed system. These functions remain outside the scope of reservation management and may be addressed in future system enhancements.

the proposed system does not include payroll processing, employee attendance monitoring, human resource management, recruitment management, employee performance evaluation, or personnel administration features. The study is focused exclusively on reservation-related operations and does not attempt to automate other administrative functions of the resort.

The system does not provide customer self-service reservation capabilities through public online registration portals. The primary users of the system are resort administrators and authorized staff members responsible for managing reservations. Customers will not directly interact with the system for account creation, online booking management, or profile maintenance. Any customer interactions remain subject to existing resort procedures.

The proposed system does not include electronic document management functions such as digital contract signing, electronic signatures, automated legal documentation, or regulatory compliance management. The study concentrates only on reservation processing and information management requirements relevant to the resort's operations.

1. Online payment processing or integration with payment gateways, which remain under the resort's existing procedures.
2. Native mobile application development. The interface may be responsive, but the primary platform is desktop-based.
3. Integration with external booking platforms and travel agencies.
4. Customer self-service portals. Guests do not create accounts or manage bookings directly in the system.
5. Reservation of other facilities such as function halls, cottages, and swimming pool packages.
6. Automated SMS, email, or push notifications.
7. Inventory, payroll, human resource, restaurant, accounting, and other management functions. The system is not a complete resort management information system.
8. Advanced security technologies such as multi-factor authentication, biometric authentication, intrusion detection and prevention systems, and SIEM platforms.
9. Penetration testing, vulnerability scanning, and digital forensics. Security evaluation is limited to authentication, access control, password protection, and activity logging.
10. Cloud infrastructure management, multi-branch deployment, and enterprise-level disaster recovery.

11. Artificial intelligence, predictive analytics, and other advanced business intelligence tools.
12. Large-scale customer surveys. Evaluation focuses on system functionality, usability, efficiency, reliability, and security, and is conducted with selected resort staff and administrators within a limited development period.

The system supports management decisions but does not replace the authority of resort management over room allocation, reservation approval, and operational policies. It also does not guarantee the complete elimination of all errors or security risks, since system effectiveness depends on proper user training, responsible use, and regular maintenance.

1.9 Target Users

The primary target users of the proposed web-based reservation management system are the management and authorized personnel of Coravergel Inland Resort. These users are directly involved in reservation processing, guest management, room allocation, operational monitoring, and administrative activities related to the resort's accommodation services. The system is specifically designed to support their daily responsibilities by providing a centralized, secure, and efficient platform for managing reservation-related information.

The system is intended for internal use by the following authorized personnel of Coravergel Inland Resort:

1. **System administrators** Administrators have the highest level of access within the system and are responsible for overseeing its overall operation and maintenance. Their responsibilities include managing user accounts, assigning user roles and permissions, monitoring system activities, reviewing audit logs, generating reports, maintaining data accuracy, and ensuring the security of reservation records. Administrators are also responsible for controlling access to sensitive information and implementing security measures that protect customer data from unauthorized access or misuse. Through the system's administrative functions, they can effectively supervise reservation operations and ensure compliance with organizational policies.
2. **Receptionists and reservation staff** These personnel are responsible for the day-to-day operation of the reservation process. Their duties include creating new reservations, updating booking information, checking room availability, processing cancellations, modifying guest records, and responding to customer inquiries. The proposed system is intended to simplify these tasks by providing a user-friendly interface that enables faster processing of reservations and more accurate management of customer information. By

reducing manual paperwork and repetitive administrative tasks, the system helps staff members improve productivity and service efficiency.

3. **Front desk personnel** who regularly interact with guests and require immediate access to reservation information. These users can benefit from the real-time availability monitoring features of the system, allowing them to quickly verify room status and reservation details. Access to updated information enables front desk staff to provide better customer service, reduce waiting times, and improve the overall guest experience.
4. **Resort managers and supervisors** who require access to operational reports and reservation summaries for decision-making purposes. Through the system's reporting features, managers can monitor reservation trends, occupancy rates, room utilization, and staff activities. These reports provide valuable information that can assist management in evaluating business performance, planning operational improvements, and allocating resources more effectively. Access to accurate and timely information supports informed decision-making and strategic planning.
5. **IT or designated technical personnel** designated technical staff responsible for maintaining the system infrastructure. Although their interaction with reservation functions may be limited, they may assist in system maintenance, troubleshooting, database management, user account recovery, software updates, and security monitoring. Their involvement helps ensure that the system remains functional, secure, and available for authorized users **.**
6. **Resort owners** while they may not use the system on a daily basis, they can gain valuable insights from generated reports and operational data. Improved reservation management, increased efficiency, reduced booking errors, and enhanced data security contribute to better business performance and customer satisfaction, which ultimately support the long-term success of the resort.

Although customers and guests are important stakeholders in the reservation process, they are not considered direct users of the system within the scope of this study. The proposed reservation management system is intended primarily for internal operational use by authorized resort personnel. Guests will continue to communicate with the resort through existing reservation channels, while resort staff will process and manage reservations using the system. Therefore, customers will not have direct access to create accounts, modify reservations, or access reservation records within the system.

In terms of cybersecurity, all target users are expected to follow established security procedures when accessing the system. User authentication mechanisms, password protection, role-based access control, and activity monitoring features are implemented to ensure that each user can only access information and functions appropriate to their

assigned responsibilities. These security controls help protect sensitive guest information while maintaining accountability among system users.

The system is designed to accommodate users with varying levels of technical expertise. Its interface and functionalities are developed to be simple, intuitive, and easy to understand, allowing staff members to perform their tasks efficiently even with limited technical knowledge. This user-centered design approach helps reduce training requirements and encourages effective system adoption within the organization.

the target users of the proposed web-based reservation management system include administrators, resort managers, supervisors, receptionists, reservation staff, front desk personnel, and authorized technical personnel of Coravergel Inland Resort. By addressing the needs of these users, the system aims to improve operational efficiency, strengthen information security, enhance reservation management, and contribute to the overall quality of service provided by the resort.

1.10 Platform of the Study

The proposed system is a web-based application that runs in modern browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge, so no specialized software needs to be installed on staff computers. It follows a client-server architecture in which the browser handles the interface and user input, while the server handles business logic and communicates with a centralized database that stores reservations, room information, guest profiles, user accounts, activity logs, and reports.

The proposed platform is designed to operate within the resort's existing network infrastructure. Depending on organizational requirements, the system may function through a Local Area Network (LAN) environment, allowing authorized personnel to access the system within the resort premises. Alternatively, the system may be deployed on an internet-accessible web server, enabling remote access for authorized administrators when necessary. This flexibility allows the resort to select a deployment strategy that best fits its operational requirements and technical resources.

To support information security, the platform incorporates secure user authentication and authorization mechanisms. Authorized users must provide valid credentials before accessing system functions. User roles and permissions are managed through role-based access controls that limit access according to specific responsibilities. This security structure helps protect sensitive reservation and customer information while preventing unauthorized activities within the system.

The platform is developed using modern web development technologies that support reliability, maintainability, and scalability. The front-end component is responsible for creating an interactive and user-friendly interface, while the back-end component handles data processing, transaction management, and communication with the database. These technologies work together to provide a responsive and efficient system environment for daily reservation operations.

The platform supports real-time information processing, allowing updates made by one authorized user to become immediately available to other users within the system. This capability is particularly important for room availability monitoring, reservation modifications, booking confirmations, and guest record management. Real-time data synchronization reduces communication delays and minimizes the possibility of conflicting reservation information.

The system is also designed with usability and accessibility in mind. The user interface follows a structured layout that enables users to navigate system functions easily and efficiently. Menu options, reservation forms, search tools, and reporting features are organized to support quick access to information and reduce the learning curve for new users. This design approach contributes to improved productivity and a better overall user experience.

The system can operate on the resort's local area network or on an internet-accessible server for authorized remote access, depending on the resort's needs and resources. Updates made by one authorized user are immediately visible to other users, which supports real-time room monitoring. Security is built into the platform through authentication, role-based access control, and activity logging, consistent with the recommendation to embed privacy and security in the system design lifecycle [8]. Database backup and recovery support protect records from accidental loss. The primary platform is desktop and laptop computers used by resort staff.

1.11 Significance of the Study

This study on the development of a secure, efficient, and centralized web-based reservation system for Coravergel Inland Resort holds significant value to different individuals and groups, particularly in improving reservation management, strengthening data security, and enhancing overall operational performance within the resort. As the hospitality industry continues to adopt modern technologies to improve service quality and operational effectiveness, the need for a reliable and secure reservation management system becomes increasingly important. The proposed system addresses the limitations of traditional manual reservation methods by introducing automation, centralized data management, real-time information processing, and cybersecurity measures that support more effective business operations

This study is significant to the following groups:

Resort owner and management.

The system gives management one organized source of reservation data. They can monitor bookings and occupancy in real time, review who performed each action through audit logs, and use reports for planning. These benefits are consistent with research showing that digital technologies improve hotel service quality, resource utilization, and competitiveness [1], [2]. Stronger access control and logging also help management protect guest data, which is a recognized priority for hospitality businesses [6], [7].

The proposed system helps administrators reduce the operational challenges associated with manual reservation management. Since all reservation records are stored in a centralized database, management can quickly access information whenever needed without spending excessive time searching through paper documents, notebooks, or multiple spreadsheets. This improves administrative efficiency and enables managers to focus on strategic planning and business development rather than routine record-keeping tasks.

The system also provides management with a clearer overview of the resort's accommodation operations. Through real-time monitoring of room availability and reservation status, administrators can immediately identify occupancy trends, booking patterns, and room utilization rates. This information can help management optimize room allocation, maximize occupancy levels, and improve overall business performance.

Resort staff.

Automated booking, instant room availability, and quick record search reduce paperwork, repeated encoding, and double bookings. Studies of web-based reservation systems report shorter processing time and fewer recording errors compared with manual methods [3], [4]. Role-based access also clarifies which records each staff member may view or modify.

The system also improves coordination among employees by ensuring that all reservation data is stored in one centralized database, making it easier for staff to access and update information whenever needed. Since all authorized personnel can view the same updated records, communication gaps and misunderstandings regarding room status, reservation schedules, and guest information are significantly reduced. This promotes teamwork and ensures that employees can work together more effectively in managing resort operations.

The proposed system helps staff save valuable time when searching for reservation records and guest information. In a manual environment, employees often spend considerable time locating documents, reviewing reservation logs, and verifying booking details. With the implementation of a centralized digital system, information can be retrieved

instantly through search and filtering functions. This allows staff members to respond more quickly to customer inquiries and provide more efficient service.

The resort as a whole.

Faster and more accurate service supports guest satisfaction and trust. A centralized, backed-up database improves business continuity, and digital records reduce reliance on paper. The system also supports the resort's compliance with the Data Privacy Act of 2012 by applying security measures to the personal data it collects [8].

The system also enhances data security by protecting guest information from loss, unauthorized access, and data manipulation. With secure storage and controlled access, the resort can maintain the confidentiality and integrity of customer data, which is essential for maintaining trust and credibility. Security features such as user authentication, role-based access control, password protection, and activity monitoring help safeguard sensitive information while ensuring that only authorized personnel can access reservation records.

The improved efficiency and accuracy of the reservation system directly contribute to customer satisfaction. Guests are less likely to experience issues such as booking errors, delays, or misunderstandings. This leads to better customer experiences, improved trust in the resort, and a stronger reputation in the hospitality industry. Satisfied customers are more likely to return for future visits and recommend the resort to friends, family members, and other potential guests, contributing to business growth and customer retention.

The proposed system also improves the overall management of reservation data by providing a centralized database where all information is stored, updated, and managed efficiently. Instead of maintaining multiple notebooks, spreadsheets, or paper files, the resort can access all reservation information through a single platform. This centralized approach improves data consistency, reduces information duplication, and minimizes the risk of lost or misplaced records.

Future researchers.

The study provides a reference for reservation systems that combine automation with cybersecurity controls. It shows how a rapid or iterative development approach, user authentication, role-based access control, and audit logging can be applied in a small hospitality setting. Future work may extend it with multi-factor authentication, cloud deployment, notifications, or analytics.

Future researchers can utilize this study as a guide in understanding system design principles, software development life cycle models, and development methodologies such as Rapid Application Development (RAD) or similar

iterative approaches. These methodologies highlight the importance of continuous user feedback, prototyping, and incremental improvements, which are essential in building effective and user-centered systems.

This research contributes practical knowledge regarding the implementation of essential system functionalities such as user authentication, role-based access control, reservation management, and activity logging. These features are crucial in ensuring system reliability, preventing unauthorized access, and maintaining the integrity and confidentiality of user data. Researchers may further explore how these security mechanisms can be enhanced using modern technologies such as multi-factor authentication and encryption techniques.

This study may serve as a foundation for exploring more advanced system enhancements. Future developments may include mobile application integration, real-time booking notifications, automated confirmation systems, cloud-based database management, and integration with third-party platforms. These improvements can significantly increase system scalability, usability, and overall performance in handling large volumes of users and transactions.

Developers and IT practitioners.

This research provides insights into key development practices such as modular system design, data normalization, and the implementation of secure authentication and authorization processes. These practices are essential in ensuring system stability, maintainability, and data integrity. IT practitioners may use these concepts to guide the development of scalable systems that can handle increasing user demands over time.

developers can enhance or expand the system by incorporating advanced functionalities such as online payment gateway integration, mobile application compatibility, automated email and SMS notifications, and third-party booking platform synchronization. These improvements can significantly increase user convenience and expand the system’s usability in real-world applications.

Future enhancements may also include cloud-based deployment to improve accessibility and system reliability, as well as the integration of analytics dashboards to monitor reservation trends and system performance. By adopting these advancements, IT practitioners can further optimize system efficiency and deliver more innovative solutions within the hospitality industry.

1.12 Definition of Terms

To ensure clarity and better understanding of the study, the following terms are defined based on how they are used in the development of the web-based reservation system for Coravergel Inland Resort. These definitions are

provided to establish a common understanding of important technical and conceptual terms used throughout the study.

Each term is explained in the context of system development, functionality, and its relevance to the research. This section helps eliminate ambiguity, allowing readers, researchers, and system users to clearly understand how each concept is applied within the system. It also serves as a guide for interpreting technical processes, system features, and security mechanisms involved in the development and implementation of the reservation system.

The definitions are tailored specifically to the scope of the study, particularly in relation to web-based applications, database management, user roles, and system security. By clearly defining these terms, the study ensures consistency in interpretation and strengthens the overall comprehension of the system's design, structure, and operational processes. This section also supports the academic integrity of the research by providing standardized meanings for key concepts used throughout the documentation.

The following terms are defined as they are used in this study.

Access control

Access Control refers to a security mechanism used in the system that restricts and regulates who can view, modify, or manage specific data and features within the reservation system. It is an essential part of system security that determines the level of access granted to different users based on their roles and responsibilities. Access control ensures that only authorized users such as administrators and designated staff are allowed to perform sensitive actions like editing reservations, deleting records, or viewing confidential guest information. By implementing access control, the system reduces the risk of unauthorized operations, prevents data misuse, and ensures that users can only interact with features relevant to their roles. This helps maintain data integrity, confidentiality, and overall system reliability

Audit trail (activity log) is the record of user actions in the system, such as logins and the creation, update, or deletion of reservations.

Authentication

Authentication is the process of verifying the identity of a user before allowing access to the system. It is typically implemented using login credentials such as usernames and passwords, and serves as the first layer of system security authentication ensures that only registered and verified users can access the reservation platform. This process protects the system from unauthorized entry and potential security breaches. By requiring users to prove

their identity before gaining access, authentication strengthens system security and ensures that all actions performed within the system are traceable to legitimate users.

Backup and recovery

Refers to the process of creating copies of important system data and restoring those copies when the original data is lost, damaged, accidentally deleted, or affected by system failure. Backup involves regularly saving copies of information in a secure storage location, while recovery involves retrieving and restoring the saved data to help return the system to normal operation. This process is an essential part of data protection because it reduces the risk of permanent data loss and helps maintain business continuity.

In the web-based reservation management system for Coravergel Inland Resort, backup and recovery can be applied by regularly creating copies of the centralized database containing guest information, reservation records, room availability details, user accounts, and activity logs. These backups may be stored in a secure external storage device or another protected location. If the database becomes corrupted, files are accidentally deleted, or a technical problem causes data loss, the administrator can restore the most recent valid backup to recover the affected records.

Booking

Booking refers to the process of reserving a room or service in advance through the reservation system. It is one of the core functions of the system and represents the actual transaction between the user and the resort booking involves entering customer details, selecting room types, choosing reservation dates, and confirming availability through the system. The automated booking process helps eliminate manual encoding errors, speeds up transaction handling, and ensures real-time accuracy of reservation data. This results in a more efficient and reliable reservation process for both staff and customers.

Centralized system

A Centralized System is a system architecture where all data and information are stored, managed, and processed in a single central database or server. This structure allows for unified data management and consistent access across all authorized users. The centralized system ensures that all reservation records, guest information, and transaction logs are stored in one secure platform. This design improves data consistency, reduces duplication, and minimizes the risk of data fragmentation. It also makes it easier for administrators to manage, update, and retrieve information efficiently, supporting smoother system operations.

CIA triad

refers to the three fundamental principles of information security that help protect data and ensure the reliability of an information system. These principles guide the implementation of security measures to prevent unauthorized access, maintain accurate information, and ensure that data and services remain accessible to authorized users when needed.

In the web-based reservation management system for Coravergel Inland Resort, confidentiality ensures that sensitive information, such as guest details, reservation records, and user credentials, can only be accessed by authorized personnel. Integrity ensures that reservation records, guest information, and room availability details remain accurate, complete, and protected from unauthorized modification or deletion. Availability ensures that the system and its stored information are accessible to authorized resort personnel whenever needed to manage reservations and check room availability.

Data security

Data Security refers to the protection of digital information from unauthorized access, alteration, destruction, or loss. It is a critical component of any information system that handles sensitive data. The data security ensures that all guest details and reservation records are protected through secure login systems, controlled access levels, and safe data storage practices. The system aims to preserve the confidentiality, integrity, and availability of data by preventing unauthorized access and reducing the risk of data breaches or system manipulation.

Real-time tracking

Real-Time Tracking refers to the system's ability to instantly update and display changes as they occur. It ensures that users always have access to the most current information without delay. The real-time tracking allows staff to immediately view updated room availability, reservation status, and booking changes. This feature is essential in preventing overbooking and ensuring that all transactions are accurately reflected in the system at the moment they occur.

Reservation

A Reservation refers to an arrangement made in advance to secure a room or service for a specific date and time. It represents the core transaction processed by the system. Reservations include customer information, booking schedules, selected rooms, and status updates stored in the system database. Each reservation is systematically recorded to ensure proper tracking, organization, and management of all bookings within the resort.

Digital Reservation System

A Digital Reservation System is a computerized platform designed to manage booking transactions electronically without the need for manual processes. It automates reservation handling and improves operational efficiency The digital reservation system refers to the web-based platform developed for Coravergel Inland Resort. It replaces traditional manual methods by automating bookings, storing data digitally, and providing real-time updates on room availability and reservation status. This system improves speed, accuracy, and convenience for both staff and guests.

Manual Reservation System

A Manual Reservation System is a traditional method of handling bookings using paper records, logbooks, or basic spreadsheets. It relies heavily on human input and physical documentation.

The manual reservation system refers to the previous process used by the resort before automation. This method is prone to human errors, such as double booking, data loss, and inconsistent records. It is also time-consuming and inefficient, especially when handling multiple reservations simultaneously.

Role-based access control (RBAC)

A security mechanism that manages access to a system by assigning specific permissions to users based on their roles and responsibilities. Instead of allowing every user to access all system features, RBAC ensures that each user can only perform actions authorized for their assigned role. This helps protect sensitive information, prevent unauthorized access, and maintain the security and integrity of system data.

In the web-based reservation management system for Coravergel Inland Resort, RBAC can be implemented by assigning different permissions to the administrator and reservation staff. The administrator may have permission to manage user accounts, update room information, view all reservation records, and monitor system activity logs. Meanwhile, reservation staff may be permitted to create, view, update, and manage guest reservations and check room availability, but they may not be allowed to change security settings or manage user accounts.

User Interface (UI)

User Interface refers to the visual components of the system that users interact with, such as buttons, forms, menus, dashboards, and navigation elements.

The UI is designed to be simple, organized, and user-friendly to ensure that staff and administrators can easily operate the system. A well-designed interface improves usability, reduces user confusion, and enhances the overall efficiency of the reservation process.

User Experience (UX)

User Experience refers to the overall satisfaction and ease of use experienced by users when interacting with the system. It includes factors such as system speed, usability, accessibility, and overall efficiency. The UX focuses on ensuring that the reservation system is intuitive, responsive, and convenient for users. A positive user experience helps reduce operational difficulties, improves workflow efficiency, and ensures that users can complete tasks quickly and accurately without complications.

REFERENCE

[1] M. Iranmanesh, M. Ghobakhloo, M. Nilashi, M.-L. Tseng, E. Yadegaridehkordi, and N. Leung, "Applications of disruptive digital technologies in hotel industry: A systematic review," *Int. J. Hospitality Manage.*, vol. 107, Art. no. 103304, Oct. 2022, doi: 10.1016/j.ijhm.2022.103304.

[2] J. Marques and R. P. Marques, Eds., *Digital Transformation of the Hotel Industry: Theories, Practices, and Global Challenges*. Cham, Switzerland: Springer, 2023, doi: 10.1007/978-3-031-31682-1.

[3] E. M. Benolon, Y. P. K. Kelen, and S. S. Manek, "Design and development of Ariesta Hotel reservation information system based website using waterfall method," *J-ENSITEC (J. Eng. Sustain. Technol.)*, vol. 12, no. 2, pp. 10497–10507, Jun. 2026, doi: 10.31949/j-ensitec.v12i02.18002.

[4] Nurhaedar and L. Mariana, "Implementasi reservasi kamar berbasis web pada Hotel Bintang Wisata Pantai Galesong Kabupaten Takalar," *JUMISTIK*, vol. 1, no. 1, 2022, doi: 10.70247/jumistik.v1i1.5.

[5] Y. Adams, "Design and implementation of a web based reservation system for a hospitality industry in Paragon Hotel," *Newport Int. J. Eng. Phys. Sci.*, 2023. [Online]. Available: https://kiu.ac.ug/drice/publications/702/design-and-implementation-of-a-web-based-reservation-system-for-a-hospitality-industry-in-paragon-hotel

[6] M. Pragnell and J. Farley, "Hotel industry cyber update – September 2022," Arthur J. Gallagher & Co., 2022. [Online]. Available: https://www.ajg.com/news-and-insights/hotel-industry-cyber-update-september-2022/

[7] K. Pipyros and S. Liasidou, "A new cybersecurity risk assessment framework for the hospitality industry: Techniques and methods for enhanced data protection and threat mitigation," *Worldwide Hospitality Tourism Themes*, vol. 17, no. 1, pp. 48–61, Mar. 2025, doi: 10.1108/WHATT-12-2024-0296.

[8] D. Cheng, "Managing data privacy compliance and technology rollout in the Philippines post COVID-19," in *Proc. De La Salle Univ. Res. Congress*, 2023. [Online]. Available: https://www.dlsu.edu.ph/wp-content/uploads/pdf/conferences/research-congress-proceedings/2023/HCT-01.pdf

[9] N. H. Hishamudin and N. A. Abdullah, "Secured restaurant reservation system using role-based access control approach," Appl. Inf. Technol. Comput. Sci., vol. 3, no. 2, pp. 165-184, Nov. 2022, doi: 10.30880/aitcs.2022.03.02.011.

[10] Y. Adams, "Design and implementation of a web based reservation system for a hospitality industry in Paragon Hotel," Newport Int. J. Eng. Phys. Sci., 2023.

[11] OWASP Foundation, "OWASP Top 10:2021," Sep. 2021. [Online]. Available: https://owasp.org/Top10/2021/

[12] OWASP Foundation, "OWASP Top 10:2025," 2025. [Online]. Available: https://owasp.org/Top10/2025/

[13] D. Temoshok et al., "Digital identity guidelines," National Institute of Standards and Technology, Gaithersburg, MD, USA, NIST Special Publication 800-63-4, Jul. 2025, doi: 10.6028/NIST.SP.800-63-4.

[14] National Institute of Standards and Technology, "The NIST cybersecurity framework (CSF) 2.0," NIST, Gaithersburg, MD, USA, Cybersecurity White Paper 29, Feb. 2024, doi: 10.6028/NIST.CSWP.29.

[15] Systems and Software Engineering - Systems and Software Quality Requirements and Evaluation (SQuaRE) - Product Quality Model, ISO/IEC Standard 25010:2023, 2023.

[16] Information Security, Cybersecurity and Privacy Protection - Information Security Management Systems - Requirements, ISO/IEC Standard 27001:2022, 2022.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Foreign Studies

Foreign studies on hotel, resort, and related reservation systems show a steady shift toward web-based, centralized, and security-aware platforms. Recent work covers system design and implementation, user acceptance of online and mobile booking, security controls such as authentication and access control, and the use of automation, artificial intelligence (AI), and analytics in hospitality operations.

Web-based reservation system design. Adams [1] designed and implemented a web-based reservation system for the Paragon Hotel, using HTML, PHP, and JavaScript for the front end and MySQL for the back end. The system maintains guest and room records, including room categories and their corresponding charges. It separates user roles: guests make reservations online, receptionists view all reservations and may send feedback to guests, and administrators can view, delete, and assign reservations. The study also examined how guests reach a booking decision, treating the reservation as the end result of a decision process rather than only a database transaction. Its role-based structure and use of a centralized relational database are directly relevant to the system proposed for Coravergel Inland Resort.

Security in reservation systems. Hishamudin and Abdullah [2] developed a secured restaurant reservation system in Malaysia. They noted that existing reservation systems may be exposed to unauthorized access and fraudulent activities that can lead to information theft. To address this, their system applied a password policy for sign-in, two-factor authentication, and role-based access control (RBAC), and was built with PHP using a prototype model. They reported that the system prioritized confidentiality and integrity by limiting unauthorized access to its database. Although the study concerns restaurants, its security design transfers readily to resort reservation platforms that store guest information.

Information technology and hotel performance. Ezzaouia and Bulchand-Gidumal [3] examined how information technology (IT) adoption affects hotel performance in Morocco, a developing country, using responses from 100 general managers of three- to five-star hotels. Their results also revealed a significant relationship between employee

performance and financial performance in hotels. This supports the view that technology adopted in a developing-country setting is relevant not only to operations but also to staff productivity and the financial results of the business.

User acceptance of online and mobile booking. Ali et al. [4] compared hotel websites and hotel mobile apps using the stimulus-organism-response framework, with 257 website users and 292 mobile app users. They found that the quality of both websites and apps positively influenced telepresence, flow, and engagement, and that telepresence and flow positively affected booking intentions in both samples. For website users, however, engagement did not significantly affect booking intentions. These results suggest that the quality of the booking interface is itself a factor in whether guests complete a reservation, and that a reservation platform should be judged by usability as well as by features.

Contactless and digital operations after the pandemic. A 2022 hotelier technology sentiment survey by Stayntouch and the NYU Tisch Center reported that technology spending by hotels rose by 31 percent in under two years, and that adoption of contactless technology (self-service check-in, in-room technology, mobile keys, and digital payments) increased by 66 percent during the pandemic [5]. Nearly 75 percent of respondents believed that contactless technology would become a long-term trend [5]. This indicates that the move toward online and automated guest transactions is likely to continue rather than reverse.

AI and chatbots in hospitality. Sam and Jasim [6] reviewed 48 studies from 2019 to 2023 on chatbot use in hotels, restaurants, and tourism, and noted that research on chatbots specifically in restaurant and hotel contexts is still scarce. Camilleri and Troise [7] similarly reviewed AI customer-service chatbots in travel and hospitality and concluded that tourism businesses still have room to improve the efficiency and responsiveness of their online customer service, but that chatbots cannot yet replace human agents in all interactions because they cannot resolve complex queries and complaints. Together, these studies indicate that AI-assisted features may complement a reservation system but should not be treated as a substitute for staff.

Analytics and demand forecasting. Henriques and Pereira [8] conducted a systematic review of 20 studies on AI-based hotel demand forecasting and concluded that AI methods can significantly improve forecasting accuracy and support data-driven decisions, while noting the need for high-quality data. Jawabreh [9] reviewed big data in the hospitality industry and reported that hotels use data on occupancy rates and current bookings, among other sources, to price and manage customers and to forecast demand levels more accurately, which in turn influences revenue management strategy. These studies show that the records produced by a reservation system have value beyond booking, provided that the data are accurate and well organized.

These findings strongly support the development of a secure and centralized web-based reservation management system for Coravergel Inland Resort because the resort currently experiences similar operational challenges identified in international studies. The proposed study aligns with global technological trends in the hospitality industry by emphasizing operational efficiency, accurate reservation handling, organized data management, customer convenience, and protection of customer information through modern web-based technologies.

2.2 Local Studies

Local studies in the Philippines regarding reservation systems in hotels, resorts, and other hospitality businesses consistently show a growing need for digital transformation in the tourism and hospitality industry. These studies reveal that many small and medium-sized establishments continue to rely on manual or semi-manual reservation systems, which often result in inefficiency, booking errors, delayed processing, and security vulnerabilities. Because of these recurring issues, local researchers strongly emphasize that the adoption of web-based reservation systems is essential to improve operational performance, enhance data accuracy, and increase customer satisfaction.

Polines et al. [10] presented a narrative literature review of studies published between 2015 and 2025 on how digital marketing strategies influence booking behavior in the Philippine hospitality sector. Guided by the Technology Acceptance Model, the Stimulus-Organism-Response framework, and Social Proof Theory, the review found that intuitive booking interfaces reduce friction, that trust (reinforced by reviews and reputation management) reduces perceived risk, and that integrated strategies across search engines, online travel agencies (OTAs), and social media yield stronger booking outcomes, including higher direct reservations and improved loyalty [10]. This supports the importance of ease of use and trust in a resort's own booking platform.

Atienza and Ylagan [11] studied digital marketing, guest experience, and behavioral intention among resorts in Calabarzon, Philippines. Their work is relevant because it focuses on resorts rather than hotels and links the digital presence of the resort to the guest experience and to the guest's intention to book. Some local studies emphasize the role of data analytics in improving hospitality management decisions. These studies explain that analyzing reservation trends, peak seasons, and customer preferences allows resort managers to optimize pricing, staffing schedules, and resource allocation. Although still emerging in local establishments, this capability demonstrates the potential of reservation systems to evolve from simple booking tools into strategic decision-support systems.

At the market level, a report on Philippine online travel notes that the country's digital economy reached USD 36 billion in 2025 and is forecast to nearly double by 2030, while international OTAs such as Agoda and Booking.com still dominate online bookings [12]. The same report describes a local provider that absorbs revenue risk so that small, unbanked island properties can accept GCash and credit card payments from guests [12]. These findings show

both the opportunity and the dependence on third-party platforms that small independent properties face, and they indicate that a resort-owned reservation system can support direct bookings and local payment practices.

2.3 Related Literature

This study focuses on the importance of information systems, web-based technologies, database management, software engineering principles, cybersecurity practices, cloud computing, and human-computer interaction in the development of modern reservation systems. These concepts are essential in building a secure, efficient, scalable, and centralized web-based reservation system for Coravergel Inland Resort. The literature also highlights how digital systems improve operational performance, reduce human error, enhance data accuracy, support real-time processing, and increase customer satisfaction in the hospitality industry.

2.3.1 Cybersecurity and Data Protection

Reservation systems store personal and sometimes financial information, which makes security a central design concern. A 2023 report by Trustwave found that 31 percent of hospitality organizations have experienced a data breach [13]. Hotels retain sensitive data such as credit card numbers, personal details, and passport numbers in their reservation databases, and the hospitality industry has been reported to have the highest phish-prone percentage of any industry, at 48 percent [14]. The same source emphasizes that human error is often the biggest issue in data breaches [14].

The Open Worldwide Application Security Project (OWASP) released its Top 10 list of web application risks for 2025, its first update since 2021 [15], [16]. Broken access control remains the top-ranked category [15], [16]. OWASP's co-leads describe it as including violations of the principle of least privilege and APIs with missing access controls, and recommend denying access by default except for public resources [16]. The 2025 list also retains categories for authentication failures and for logging and alerting failures [15]. These categories correspond to the controls that a reservation system should provide: authentication, role-based authorization, and activity logging.

2.3.2 Access Control and Authentication

Role-based access control restricts what each user may view or change according to the user's role. Hishamudin and Abdullah [2] demonstrated its use together with a password policy and two-factor authentication in a reservation system, and Adams [1] applied separate guest, receptionist, and administrator roles in a hotel reservation system. These studies show that separating privileges is a practical and commonly used design for reservation systems and not limited to large enterprises.

2.3.3 Usability, Interface Quality, and User Acceptance

The reviewed literature repeatedly links interface quality to booking outcomes. Ali et al. [4] found that website and mobile app quality influenced telepresence, flow, and engagement, which in turn affected booking intentions. Polines et al. [10] reported that intuitive booking interfaces reduce friction in the Philippine context. These findings support user-centered design as a requirement for both guest-facing and staff-facing parts of a reservation system.

2.3.4 Automation, AI, and Contactless Services

Digital check-in, digital payments, and other contactless services grew substantially during the pandemic and are expected by many hoteliers to continue [5]. AI chatbots are increasingly studied as a way to answer guest inquiries, but current evidence suggests they are limited in handling complex queries and complaints [6], [7]. For a small resort, automation is therefore most defensible in routine, rule-based tasks such as booking validation, confirmation, and availability updates, with staff retained for exceptions.

2.3.5 Data Management, Analytics, and Decision Support

Hospitality researchers describe booking and occupancy data as inputs to pricing, demand forecasting, and revenue management [8], [9]. Henriques and Pereira [8] caution that the benefits depend on data quality. A centralized and validated database is therefore a prerequisite for any later reporting or analytics function in a resort reservation system.

2.4 Synthesis of Related Literature

This synthesis brings together the foreign studies, local studies, and related literature reviewed for the study on a secure, centralized, web-based reservation management system for Coravergel Inland Resort. Instead of summarizing each source in turn, it groups the literature into six themes, shows where the sources agree, and identifies what they leave unanswered. The gaps then justify the design of the proposed system.

Only sources published in 2021 or later are used, and each was checked against its published abstract or record. The six themes are:

1. Digital transformation and hotel performance
2. Design and benefits of web-based reservation systems
3. Cybersecurity, access control, and data privacy
4. Usability and user acceptance

5. Data management and decision support
6. Development methodology, testing, and evaluation

2.4.1 Digital Transformation and Hotel Performance

The literature agrees that digital tools now shape how hotels and resorts perform. A systematic review of 79 articles found that disruptive digital technologies contribute to six dimensions of hotel performance: financial results, competitiveness, quality of service, resource utilization, flexibility, and innovation [1]. A book-length review of digital transformation in the hotel industry likewise compiles theory, practice, and case studies on using information and communication technology to improve hotel management [2]. Reservation systems sit at the center of this shift because they connect guest interaction, room allocation, and record keeping. These sources describe the industry as a whole, so they establish why a resort should digitize but do not show how a small resort should build a reservation system.

2.4.2 Design and Benefits of Web-Based Reservation Systems

System-development studies converge on one pattern: a manual process is replaced by a centralized web application with a shared database. A reservation system for a hotel used HTML, PHP, and JavaScript with a MySQL back end, kept guest and room records including room categories and charges, and separated three roles: guests reserve online, receptionists view all reservations, and administrators view, delete, and assign them [3]. A reservation system for a garden resort that had struggled with a manual process added real-time booking and digital payments to remove operational inefficiencies [4].

The two studies agree on the core design, which is a central database, role separation, and live booking information. They differ in emphasis: one stresses role structure and the guest's decision process, the other stresses real-time booking and payment. Neither, in the abstracts reviewed, foregrounds security controls such as audit logging, which is the gap this study addresses.

2.4.3 Cybersecurity, Access Control, and Data Privacy

Security is the point on which the literature is least developed for reservation systems, yet the sources that address it agree on the main controls. Interviews with hotel managers led to the conclusion that hotels must prioritize cybersecurity by developing risk assessment methods and applying technical and organizational measures [5]. In a system-level study, a restaurant reservation system noted that existing systems can expose data to unauthorized access and fraud, and it responded with a password policy, two-factor authentication, and access control, built in

PHP with a prototype model. The authors reported that limiting access to the database protected confidentiality and integrity [6].

The OWASP Top 10:2021 gives a common vocabulary for these concerns. It ranks broken access control first, cryptographic failures second, and injection third, and it also lists identification and authentication failures and security logging and monitoring failures [7]. These categories map directly to the controls a reservation system should provide: authentication, role-based authorization, password protection, input validation, and activity logging.

For a Philippine resort, privacy law adds a further requirement. A 2023 study of data privacy compliance in the Philippines after COVID-19 recommended applying Privacy by Design so that privacy and compliance are built into technology as it is rolled out [8]. Taken together, these sources support designing security and privacy into the system from the start and not adding them afterward.

2.4.3 Usability and User Acceptance

A second, smaller body of work links interface quality to booking outcomes. A comparison of 257 hotel-website users and 292 hotel-app users found that the quality of both channels positively influenced telepresence, flow, and engagement, and that telepresence and flow positively affected booking intentions [9]. The study concerns guests and not staff, but its message transfers: a booking platform succeeds partly because it is easy and pleasant to use. For a resort whose staff may have limited technical training, this supports a simple layout, clear labels, and short reservation steps.

2.4.4 Data Management and Decision Support

Reservation records are valuable beyond the booking itself. A systematic review of 20 studies on artificial intelligence in hotel demand forecasting concluded that AI methods can significantly improve forecasting accuracy and support data-driven decisions, and it also warned that results depend on high-quality data [10]. The implication for this study is practical. Analytics and forecasting are outside the study's scope, but a centralized, validated database is the precondition for any later reporting or analysis, because duplicate or incomplete records would undermine it.

2.4.5 Development Methodology, Testing, and Evaluation

Recent booking-system projects frequently choose Rapid Application Development (RAD). A ticket-booking system built to replace a manual registration process combined structured techniques, prototyping, and joint application development to speed up development [11], and a web-based registration platform used RAD for its

accelerated cycles of iterative prototyping and continuous stakeholder engagement [12]. These choices match a small project in which staff can review prototypes throughout.

On security within the development life cycle, NIST notes that few SDLC models explicitly address software security in detail, so secure development practices must be added to whichever model is used [13]. The OWASP Application Security Verification Standard supplies a checklist of requirements for verifying web application controls [14].

For testing, an evaluation of 75 real-world web applications with four static and five dynamic analysis tools found that no single tool or method detects vulnerabilities comprehensively, and that combining the approaches improves coverage [15]. This supports pairing automated scanning with manual test cases. For evaluation, ISO/IEC 25010:2023 defines a product quality model of nine characteristics that can guide acceptance criteria and measures [16].

2.3.6 Technology Adoption in Developing-Country Hospitality

Ezzaouia and Bulchand-Gidumal [3] studied IT adoption in a developing country and related it to hotel performance, while the Philippine sources point to continued reliance on third-party OTAs and to payment-access limits among small properties [10], [12]. Taken together, this literature supports solutions that are affordable, simple to use, and aligned with local payment practices.

2.5 Research Gaps

Based on the review of related foreign and local studies, existing reservation systems, and relevant literature in the hospitality industry, several research gaps have been identified that justify the development of a secure, centralized, and web-based reservation management system for Coravergel Inland Resort. Although existing reservation systems have introduced digital technologies to improve booking efficiency, many small and medium-sized hospitality establishments continue to experience challenges involving manual record-keeping, inaccurate room availability, fragmented data storage, inadequate security, and limited usability. These challenges highlight the need for a reservation management system designed around the operational requirements and available resources of a small inland resort.

One significant gap is the **Limited resort-specific reservation system studies.** The reservation systems found in the literature were designed for hotels [1] or restaurants [2]. Resorts have operational needs of their own, such as cottage

or room-type reservations, activities, and seasonal demand, and these are not directly addressed in the reviewed system studies. As discussed in the reviewed literature, smaller establishments may continue to depend on handwritten logbooks, spreadsheets, and separate digital files to manage reservations and guest information. These practices can increase staff workload, delay booking verification, and create opportunities for human error. Furthermore, the general reservation systems examined in the review do not always demonstrate how their features can be adapted to the specific workflow of a small Philippine inland resort.

Another important gap concerns **cybersecurity and data protection**. Reservation systems store sensitive guest data and the hospitality sector reports high breach exposure [13], [14]. Studies such as [2] implement security controls, but many system-development studies, such as [1], emphasize functionality and roles and report less on audit logging and security monitoring, even though logging and alerting failures and access control failures appear among the top web application risks [15]. However, some existing system-development studies emphasize booking functionality and automation without providing sufficient discussion of comprehensive security mechanisms. Weak authentication, improper access control, and insufficient activity monitoring can expose systems to unauthorized access and inappropriate modification of records**.**

The literature also reveals a gap in **real-time reservation processing and room availability tracking**. Systems that depend on manual verification or delayed record updates may display inaccurate room availability, resulting in reservation conflicts and possible double bookings. When multiple staff members rely on separate records, inconsistencies can occur between the actual room status and the information recorded in the reservation system. A centralized system with validated reservation processing is therefore necessary to improve the accuracy and consistency of room availability information also Recent Philippine sources emphasize digital marketing, booking behavior, and market conditions [10]–[12], while detailed design and implementation studies of reservation systems for small Philippine resorts are limited in the reviewed period. .

Another gap involves **centralized database management and data quality**. Manual records, spreadsheets, and disconnected digital files can produce duplicate entries, incomplete guest information, and inconsistent reservation records. Analytics and forecasting depend on high-quality data [8], [9]. Without validated, centralized records, a small resort cannot easily benefit from them. However, the practical application of these principles to the reservation workflows of small inland resorts remains an important consideration. A unified database can provide a consistent source of reservation and guest information while supporting accurate record retrieval and future reporting.

Usability and user-centered design also remain important concerns. A reservation system may provide useful features but still be difficult to operate if its interface is complicated or its workflow requires unnecessary steps. Interface quality affects booking intention [4], [10], yet small-resort staff may have limited technical training, so a

system must be simple for both staff and guests. This concern is particularly relevant to small resorts whose personnel may have limited technical experience. A system must therefore provide clear navigation, understandable labels, organized information, and straightforward reservation procedures to support efficient daily operations and reduce user errors.

Furthermore, **local evidence on reservation system design for small Philippine resorts remains limited in the reviewed literature**. The reviewed studies and sources provide information about reservation technologies, hospitality operations, digital booking practices, and related security concerns. Nevertheless, these sources do not fully establish how a secure, centralized reservation management system can be designed and implemented around the specific operational needs of Coravergel Inland Resort. Documenting the development and evaluation of a system for this establishment can contribute practical evidence regarding the application of web-based reservation technology in a small local resort.

Another identified gap concerns **the integration of reservation functions into a single system**. Small properties in the Philippines depend heavily on international OTAs for online bookings [12], which indicates a need for a resort-owned platform that supports direct reservations.When reservation management, room availability monitoring, guest information, and user access management are handled separately, staff may need to repeat tasks or verify information across multiple records. An integrated platform can connect these essential functions and reduce unnecessary manual work. Although scalability, automated communication, online payment integration, advanced analytics, mobile applications, and third-party booking integrations are relevant considerations in modern hospitality systems, they may exceed the immediate operational requirements and defined scope of the proposed project.

Overall, the reviewed literature identifies several important areas for improvement, particularly resort-specific system design, cybersecurity, real-time availability tracking, centralized data management, usability, and integrated reservation processing. These gaps establish the need for a practical reservation management solution that addresses the actual problems encountered by Coravergel Inland Resort. The proposed project focuses on these essential requirements rather than attempting to reproduce every feature of a large commercial hospitality platform.

2.5.1 How the Current Project Addresses These Gaps

The proposed web-based reservation management system for Coravergel Inland Resort addresses the identified research gaps by providing a centralized platform for managing reservation records, guest information, room availability, and authorized user activities. Unlike the existing manual process, which depends on paper-based records, notebooks, spreadsheets, or separate computer files, the proposed system organizes reservation-related

information within a centralized database. Its design prioritizes operational efficiency, data consistency, usability, and security while remaining aligned with the resort's actual requirements.

1. Resort-Specific Reservation Management

The system is designed around the actual reservation workflow of Coravergel Inland Resort, including its accommodation types, guest records, and reservation status, instead of adopting a generic hotel or restaurant model [1], [2].It focuses on essential functions relevant to the establishment, including reservation creation and management, guest information handling, and reservation status monitoring. This approach addresses the gap in applying reservation technologies to the particular needs of a small Philippine inland resort.

2. Contribution to Local System Development Evidence

The project documents the design, development, and evaluation of a reservation management system for Coravergel Inland Resort. Through this implementation, the study provides a practical example of how a small hospitality establishment can transition from manual record-keeping to a centralized web-based platform. By documenting the design, implementation, and evaluation of a reservation system for a small Philippine inland resort, the project adds practical evidence to a local area where such studies are limited [10]–[12]. .

3. Integrated Security and User Accountability

The system incorporates user authentication, role-based access control, and activity logging to protect reservation records and support accountability. Authentication verifies user credentials before access is granted, while role-based permissions restrict available functions according to assigned responsibilities. Activity logs record relevant user actions, such as creating or updating reservations, to support monitoring and investigation of record changes. The system includes authentication, role-based access control (administrator, staff, and guest roles), and activity logging, following the approaches demonstrated in [1] and [2] and the access control and logging concerns identified in the OWASP Top 10:2025 [15], [16]. .

4. Real-Time Reservation and Availability Management

The system centralizes reservation information and applies validation when processing booking transactions. Reservation creation, modification, or cancellation can update the corresponding reservation records and room availability status. By allowing authorized staff to access current information from a shared database, the system helps reduce inconsistencies, scheduling conflicts, and errors associated with manually updating separate records. The effectiveness of these functions will be assessed during system testing.

5. Centralized and Validated Database

The proposed system stores reservation records, guest information, room details, and relevant user activity records in a structured database. Required-field checks and reservation validation help prevent incomplete or invalid entries from being saved. Centralized storage reduces the need to maintain multiple independent records and supports consistent information retrieval. Also all reservation, guest, and room data are stored in a single validated database so that records are consistent and can later support reports and analytics [8], [9]. The resulting data structure can also support basic reports and future enhancements to the system.

6. User-Friendly Interface for Authorized Personnel

The system is designed with a clear layout, consistent navigation, understandable labels, and straightforward reservation procedures. These design considerations aim to make the platform easier for administrators, receptionists, and other authorized resort personnel to learn and use. By reducing unnecessary steps in routine tasks, the interface supports more efficient reservation processing and helps minimize user errors.

7. Integrated and Maintainable Web-Based Platform

The interface is designed to be simple, consistent, and responsive for both guests and staff, consistent with findings that interface quality influences booking outcomes [4], [10]. This reduces dependence on disconnected record-keeping tools and allows authorized users to work with consistent information through a shared platform. Its focused design is intended to suit the immediate needs of the resort while allowing future improvements when additional requirements are identified. The system is intended primarily for desktop and laptop browser access, with its actual accessibility depending on the deployed network and configuration.

8. Defined Scope and Future Development Opportunities

The proposed system concentrates on reservation management, guest information, room availability, authentication, access control, activity logging, and basic reporting. It does not include online payment gateway integration, direct guest self-booking, third-party booking platform integration, dedicated Android or iOS applications, or advanced analytics and forecasting; the resort-owned platform allows guests to reserve directly, reducing dependence on third-party OTAs [12]. . These exclusions maintain the project's focus on the resort's identified operational problems while recognizing possible directions for future development.

2.6 Theoretical Foundations of Entropy-Based Anomaly Detection

[17] Callegari, Giordano, and Pagano (2017) proposed an information-theoretic approach to anomaly detection that identifies abnormal behavior by examining changes in entropy within observed data. Their study explains that entropy can measure the distribution and variability of data, and significant changes in entropy can indicate patterns that differ from normal behavior. The researchers applied the approach to network traffic and found that entropy-based measurements could be used to distinguish normal from anomalous activity. This theoretical foundation can be applied to a resort reservation system. In a manual reservation environment, booking information is commonly recorded through reservation notebooks, forms, spreadsheets, or other manually maintained records. Because these records are handled manually, unusual patterns such as duplicate reservations, repeated customer information, sudden increases in bookings for a particular date, inconsistent room assignments, or unexpected cancellation patterns may be difficult to identify immediately. A reservation system incorporating entropy-based analysis could examine the distribution of reservation activities and flag substantial deviations from established normal patterns.

A related Philippine study, [18] **SORMS: Steffi's Online Reservation Management System**, specifically examined the problems associated with manual resort reservations. The researchers reported that Steffi's Garden Resort experienced operational inefficiencies under its manual reservation process and developed a web-based system to improve booking management. Their results reported fewer booking errors and faster processing compared with the manual approach. Philippine EJournals Similarly, research on the **Nature Spring Resort Web-Based Reservation and Management System** identified delayed processing, difficult reservation procedures, and the possibility of human error as problems associated with manual resort reservation processes.

2.7 Comparison of Intrusion Detection Frameworks

Table I compares the reservation architectures reviewed in this study by operational mechanism, system requirement, and key capability. The manual, paper-based process serves as the baseline. It relies on handwritten logbooks, spreadsheets, and manual encoding, needs no specialized hardware, and depends fully on staff. As a result it offers no real-time updates and carries a high risk of error, which matches the problems reported in hotels that still record bookings in logbooks [2], [4].

Architecture / Approach	Operational Mechanism	System Requirement	Key Capability

Manual / Paper-Based Reservation (Baseline)	Handwritten logbooks, spreadsheets, and manual encoding of bookings	No specialized hardware or software; fully staff-dependent	None (No real-time updates; high error and security risk)
Web-Based Client-Server Booking System [1]	Automated registration, login, and room booking through a centralized MySQL database	Standard web hosting; moderate server storage	Moderate (Reduces manual workload and double-booking)
Mobile-Responsive Reservation Platform [2]	Responsive web design optimized for smartphone and tablet browsers	Low client-side resource use; mobile browser only	High (Anytime, cross-device accessibility)
Cloud-Based Reservation System [3]	On-demand resource allocation across distributed cloud servers	Internet-dependent; recurring cloud hosting cost	High (Scalability, automatic backup, disaster recovery)
Real-Time Synchronization System [4]	Instant database updates reflected across all connected terminals	Persistent network connection; moderate server load	High (Prevents overbooking; accurate live availability)

The web-based client-server approach replaces manual recording with a centralized database. One implementation used PHP and MySQL so that receptionists could view reservations and administrators could manage them [1]. This lowers the workload and the risk of double booking. It requires only standard web hosting, so the capability gain is moderate. The mobile-responsive approach adds a layout that adapts to phones and tablets. A recent hotel system built with the Bootstrap framework reported access from multiple devices, along with shorter processing time and fewer recording errors than the manual process [2]. Cloud-based systems move storage and processing to remote servers. Vendor material on cloud property management systems points to scalability, centralized operations, and resilience as the main advantages [3]. The trade-off is dependence on an internet connection and recurring hosting

costs. Real-time synchronization updates every connected terminal immediately. It addresses the problem of room status being unavailable or out of date, which has been reported in manual hotel settings where availability could only be obtained from the receptionist [4].

Taken together, the table shows that each step away from the manual baseline improves one capability, such as automation, accessibility, scalability, or accuracy, while adding infrastructure requirements. The proposed system for Coravergel Inland Resort therefore combines a web-based client-server design with real-time synchronization. It runs on the resort's local network or a single web server, which suits the resort's limited technical resources. It also adds authentication, role-based access control, and audit logging, which the architectures in Table I do not address on their own. Note that the "Moderate" and "High" ratings are qualitative judgments, not measured results.

REFERENCE

[1] Y. Adams, "Design and implementation of a web based reservation system for a hospitality industry in Paragon Hotel," *Newport Int. J. Eng. Phys. Sci.*, 2023. [Online]. Available: https://publications.kiu.ac.ug/702/design-and-implementation-of-a-web-based-reservation-system-for-a-hospitality-industry-in-paragon-hotel

[2] N. H. Hishamudin and N. A. Abdullah, "Secured restaurant reservation system," *Appl. Inf. Technol. Comput. Sci.*, vol. 3, no. 2, pp. 165–184, Nov. 2022. [Online]. Available: https://periodical.uthm.edu.my/index.php/aitcs/article/view/7602

[3] I. Ezzaouia and J. Bulchand-Gidumal, "The impact of information technology adoption on hotel performance: Evidence from a developing country," *J. Qual. Assur. Hosp. Tour.*, vol. 24, no. 5, pp. 688–710, May 2022, doi: 10.1080/1528008X.2022.2077886.

[4] F. Ali, L. Ali, Z. Gao, A. Terrah, and G. Turktarhan, "Determinants of user's intentions to book hotels: A comparison of websites and mobile apps," *Aslib J. Inf. Manag.*, vol. 76, no. 1, pp. 16–41, 2024, doi: 10.1108/AJIM-05-2022-0239.

[5] "9 experts react to Stayntouch report finding hotel technology sentiment, and spending, are reaching a tipping point," *Hotel Online*, 2022. [Online]. Available: https://www.hotel-online.com/news/9-experts-react-to-stayntouch-report-finding-hotel-technology-sentiment-and-spending-are-reaching-a-tipping-point

[6] S. J. I. Sam and K. M. Jasim, "Diving into the technology: A systematic literature review on strategic use of chatbots in hospitality service encounters," *Manag. Rev. Q.*, vol. 75, no. 1, 2025, doi: 10.1007/s11301-023-00388-8.

[7] M. A. Camilleri and C. Troise, "Chatbot recommender systems in tourism: A systematic review and a benefit-cost analysis," in *Proc. 8th Int. Conf. Mach. Learn. Technol. (ICMLT)*, New York, NY, USA: ACM, 2023. [Online]. Available: https://www.um.edu.mt/library/oar/handle/123456789/105032

[8] H. Henriques and L. Nobre Pereira, "Hotel demand forecasting models and methods using artificial intelligence: A systematic literature review," *Tour. Manag. Stud.*, 2024, doi: 10.18089/tms.20240304.

[9] O. Jawabreh, "Big data in the hospitality industry: A methodical review," *TELKOMNIKA*, Aug. 2024. [Online]. Available: https://telkomnika.uad.ac.id/index.php/TELKOMNIKA/article/download/25543/11919

[10] N. M. M. Polines, N. D. Mendoza, J. B. Ortaleza, and R. E. F. Regidor, "Digital marketing strategies and guest booking behavior in hospitality: A narrative literature review in the Philippines," *JPAIR Multidiscip. Res. J.*, vol. 65, no. 3, 2026. [Online]. Available: https://ejournals.ph/article.php?id=35266

[11] M. M. Atienza and A. Ylagan, "Digital marketing, guest experience and behavioral intention among resorts in Calabarzon," *Int. J. Res.*, vol. 13, no. 1, 2025, doi: 10.5861/ijrsm.2025.25036.

[12] "Inside the Philippines' overlooked digital market," *PhocusWire*, 2026. [Online]. Available: https://phocuswire.com/news/online/inside-philippines-overlooked-digital-market

[13] "Report: 31% of hospitality organizations have had a data breach," *Hotel Management*, 2023. [Online]. Available: https://www.hotelmanagement.net/tech/report-31-hospitality-organizations-have-had-data-breach

[14] E. Hertzfeld, "Staff issues drive need for cybersecurity," *Hotel Management*, Jun. 2023. [Online]. Available: https://hotelmanagement.mydigitalpublication.com/articles/security

[15] OWASP Foundation, "OWASP Top 10:2025," 2025. [Online]. Available: https://owasp.org/Top10/

[16] "OWASP Top 10: Broken access control still tops app security list," *The Register*, Nov. 11, 2025. [Online]. Available: https://www.theregister.com/2025/11/11/new_owasp_top_ten_broken/

[17] —C. Callegari, S. Giordano, and M. Pagano, "An information-theoretic method for the detection of anomalies in network traffic," *Computers & Security*, vol. 70, pp. 351–365, Sep. 2017, doi: 10.1016/j.cose.2017.07.004. ScienceDirectFor the resort manual reservation study:

[18] —E. E. Balintag, K. J. H. M. Concepcion, P. J. C. Dagman, J. A. A. Ducos, D. A. Fernandez, and E. M. Villanueva, "SORMS: Steffi's Online Reservation Management System," *Psychology and Education: A Multidisciplinary Journal*, vol. 36, no. 7, pp. 783–790, 2025, doi: 10.70838/pemj.360705.

CHAPTER III

METHODOLOGY

3.1 System Development Methodology

Chosen SDLC Model: Rapid Application Development (RAD)

The study adopts the Rapid Application Development (RAD) model as the Software Development Life Cycle (SDLC) for the Web-Based Reservation System with Security Features for Coravergel Inland Resort. RAD is a user-centered approach built on iterative prototyping, continuous user participation, and frequent testing. It lets developers produce working versions quickly, collect feedback from resort staff, and refine the system in repeated cycles.

Recent reservation and booking projects have used RAD for the same reasons. A ticket-booking system for a mountain climbing site combined structured techniques, prototyping, and joint application development to speed up system development [14]. A web-based registration platform for a hospital used RAD because iterative prototyping and continuous stakeholder engagement shorten development cycles [15].

RAD suits this study for four reasons:

- **Evolving requirements.** Booking, customer records, payment recording, and reservation monitoring are processes that staff understand better once they see a working system. RAD allows earlier phases to be revisited without restarting the project.
- **Direct user involvement.** Administrators and staff review prototypes, report issues, and suggest features, which reduces misunderstandings and costly revisions after deployment.
- **Early detection of problems.** Errors and usability issues surface in each iteration, not at the end of the project.
- **Faster deployment.** Development, testing, and evaluation run in parallel, so the resort can start using the system sooner.

The model was first proposed in the 1980s to cut development time while keeping quality. It favors functional prototypes over heavy documentation, and the system matures through repeated cycles of design, test, and improvement.

RAD Compared with Waterfall, and RAD and Security

The Waterfall model completes each phase before the next begins, so requirement errors and usability problems tend to appear late and are costly to fix. A comparative case study at an insurance company reported that researchers and end users agreed that iterative, agile-style methods added value faster than the traditional Waterfall approach [16].

Other comparative work, including a survey of 145 software-industry employees, examines how practitioners perceive Waterfall and Agile when choosing a method [17]. RAD shares the iterative and feedback-driven character of agile methods while keeping a fixed sequence of planning, design, construction, and cutover. This makes it a practical fit for a small, single-site project in which the resort staff can be reached throughout.

RAD is best suited to small and medium-sized projects with committed users. A single-site reservation system for one resort fits that profile. Its success depends on staff being available to review prototypes.

Security within RAD. A main objective of this study is to protect reservation records and customer data, so security is tested in every iteration instead of added at the end. Authentication, password protection, access control, secure database management, and backups are checked and improved in each cycle. This matches NIST guidance noting that few SDLC models address software security in detail, so secure development practices must be added to whichever model is used [18]. The same guidance recommends a core set of practices that can be integrated into each SDLC implementation [18]. The web application risks to test against come from the OWASP Top 10:2021, which ranks broken access control, cryptographic failures, and injection as the leading categories [19]. The OWASP Application Security Verification Standard supplies detailed requirements for checking authentication, session management, and access control at each iteration [20].

RAD Phases at a Glance

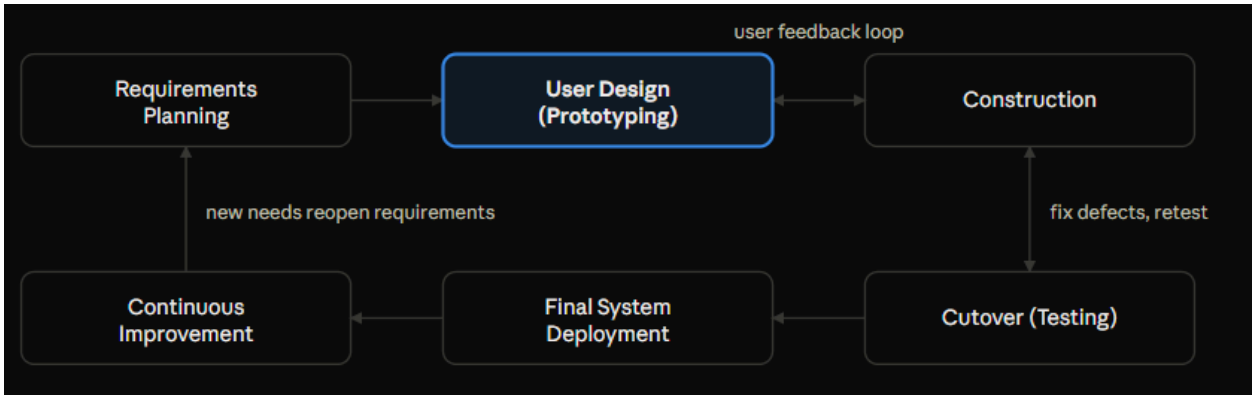


Figure 3.1

Work moves from requirements to design and construction, with users reviewing prototypes at each pass. Testing defects return to construction, and later needs to reopen the requirements.

Phases of the RAD Model

TABLE I. RAD PHASES APPLIED TO THE RESERVATION SYSTEM

Phase	Main activities	Output
1. Requirements Planning	Interviews, observation, document review, and consultation with the owner, management, and staff; identification of problems such as duplicate bookings, misplaced records, delayed confirmations, and weak data security; definition of scope, assumptions, limitations, and security requirements	Requirements specification
2. User Design (Prototyping)	Prototypes of login, reservation forms, room availability dashboard, administrative panels, database structure, and navigation; staff review and revision; DFD, ERD, flowcharts, and use case diagrams	Approved system design
3. Construction	Building and integrating the web interface, back end, database, and security features; testing each module as it is built	Working system modules
4. Cutover (Testing)	Unit, integration, and system testing, then user acceptance testing; correction of defects	Verified system
5. Final System Deployment	Installation, database configuration, account creation, data migration, final checks, and staff training	Operational system
6. Continuous Improvement	Feedback collection, maintenance, feature and security updates, and performance tuning	Sustained, evolving system

1. Requirements Planning

The researchers work with the resort owner, management, and staff to understand how reservations currently work and where they fail.

The functional requirements are customer reservation and booking management, real-time room availability monitoring, reservation approval and cancellation, user registration and authentication, report generation and reservation tracking, and administrative management tools.

The non-functional requirements are system security and data protection, reliability and accuracy of reservation information, a user-friendly interface, fast response time, access through web browsers, and database integrity with backup mechanisms.

Security requirements are fixed at this stage so that confidentiality, integrity, and availability of customer and reservation data guide the later phases.

2. User Design (Prototyping)

Staff and management interact with the prototypes, judge layout, usability, and function, and suggest changes. The prototypes are revised and may go through several iterations before the design is approved. This phase also produces the supporting diagrams: Data Flow Diagrams, Entity Relationship Diagrams, system flowcharts, and use case diagrams. The approved design becomes the basis for construction.

3. Construction

The approved design is turned into a web application. The technologies listed for this study are HTML for page structure, CSS for styling and responsiveness, JavaScript for interactivity, Python for back-end processing and business logic, and MySQL for data storage.

The main components are:

- **User Authentication Module:** registration, login, password validation, and access control.
- **Reservation Management Module:** create, update, cancel, and manage reservations.
- **Real-Time Availability Monitoring:** room status updates when bookings are made, changed, or cancelled, which limits overbooking.
- **Database Management Module:** centralized storage of customers, reservations, payments, and system logs.

- **Security and Data Protection Features:** password protection, role-based access control, secure login, backup and recovery, session management, and input validation against malicious entries.

These controls address the leading web application risks in the OWASP Top 10:2021 [19] and can be verified against the ASVS [20]. Each module is tested and corrected before it is integrated with the others.

4. Cutover: Testing and Evaluation

Testing runs at four levels, summarized in TABLE II. Defects found at any level are corrected before the next level begins.

TABLE II. TESTING LEVELS IN THE CUTOVER PHASE

Level	Focus	Examples
Unit testing	Each component on its own	Login, reservation form validation, database insertion and retrieval, availability update
Integration testing	Communication among front end, back end, and database	Login module with the authentication database; reservation module with room availability; reservation forms with the central database; report generation with stored reservation records; access control with security modules
System testing	The complete application under realistic conditions	Login, booking, availability monitoring, modification and cancellation, guest information, reports, account administration, storage and retrieval, security and access control

User acceptance testing (UAT)	Real business scenarios run by staff and administrators	Speed, accuracy, usability, and functionality judged against operational needs

Integration testing checks, for example, that a new reservation is stored correctly, that room availability updates automatically, and that the dashboard shows the change at once. It also looks for missing or duplicated data, synchronization errors, delayed status updates, and interface and back-end failures.

5. Final System Deployment

After unit, system, and user acceptance testing, the system is deployed for actual use. Deployment includes installing the system on the designated server, configuring the database and settings, creating administrator and staff accounts, migrating initial reservation and room data, running final operational checks, and confirming a stable connection between the client interface and the database server. Staff receive orientation on secure login, managing reservations, updating room availability, accessing reports, and managing guest information and user accounts, together with user manuals. The system's performance is then monitored in the working environment.

Deployment results reported by the researchers.

- Customers can reserve rooms online, which replaces handwritten records and physical logbooks.
- Staff monitor bookings on a central dashboard showing occupied, reserved, and available rooms in real time.
- Daily reservation summaries, room occupancy reports, booking history, and reservation status reports are generated automatically.
- Double bookings, incomplete records, incorrect guest information, lost documents, and delayed availability updates are minimized through validation, central storage, and real-time updates.
- Authentication, access control, and activity logging limit data access to authorized users and make staff actions accountable.
- Reservation transactions are faster, staff workload is lower, and records are better organized.

6. Continuous Improvement

RAD continues after deployment because user needs, security threats, and workloads change. Four activities keep the system current:

- **Feedback and maintenance:** collect staff suggestions on usability and workflow, fix bugs, monitor server and database performance, and update components.
- **Feature upgrades:** online payment integration, SMS or e-mail notifications, mobile-responsive improvements, advanced analytics, and a customer self-service portal.
- **Security updates:** stronger authentication and password policies, database protection updates, monitoring of suspicious activity, and tighter access control.
- **Performance improvement:** query optimization, faster response and loading, and better handling of growing reservation traffic.

3.2 System Architecture

The proposed **Web-Based Reservation System for Coravergel Inland Resort** is designed using a **three-tier centralized client–server architecture**, a widely adopted model in modern web-based information systems. This architectural design ensures efficient system processing, centralized data management, enhanced security, and real-time synchronization of reservation information among authorized users.

In this architecture, all essential system data—including reservation records, guest profiles, user credentials, room availability status, transaction history, and activity logs—are stored in a centralized server environment. Centralized storage guarantees data consistency, prevents duplication of records, and allows administrators to manage the entire reservation operation from a single secure location.

The system is accessible through standard web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. Authorized users including resort administrators and staff can access the system using desktop or laptop computers connected through a Local Area Network (LAN) or the internet. This setup provides flexibility and accessibility while maintaining administrative control over system operations.

Whenever a user performs an action such as creating a reservation, updating booking information, or checking room availability, the request is transmitted from the client interface to the application server. The server processes the request, validates inputs, enforces system rules, communicates with the database, and returns a response to the user interface. This structured communication ensures accurate processing of transactions and minimizes operational errors.

One of the primary advantages of this architecture is **real-time data processing**. All booking updates, cancellations, or room status changes are immediately reflected throughout the system. This prevents booking conflicts, eliminates manual coordination, and guarantees that all users are viewing updated and synchronized information at all times.

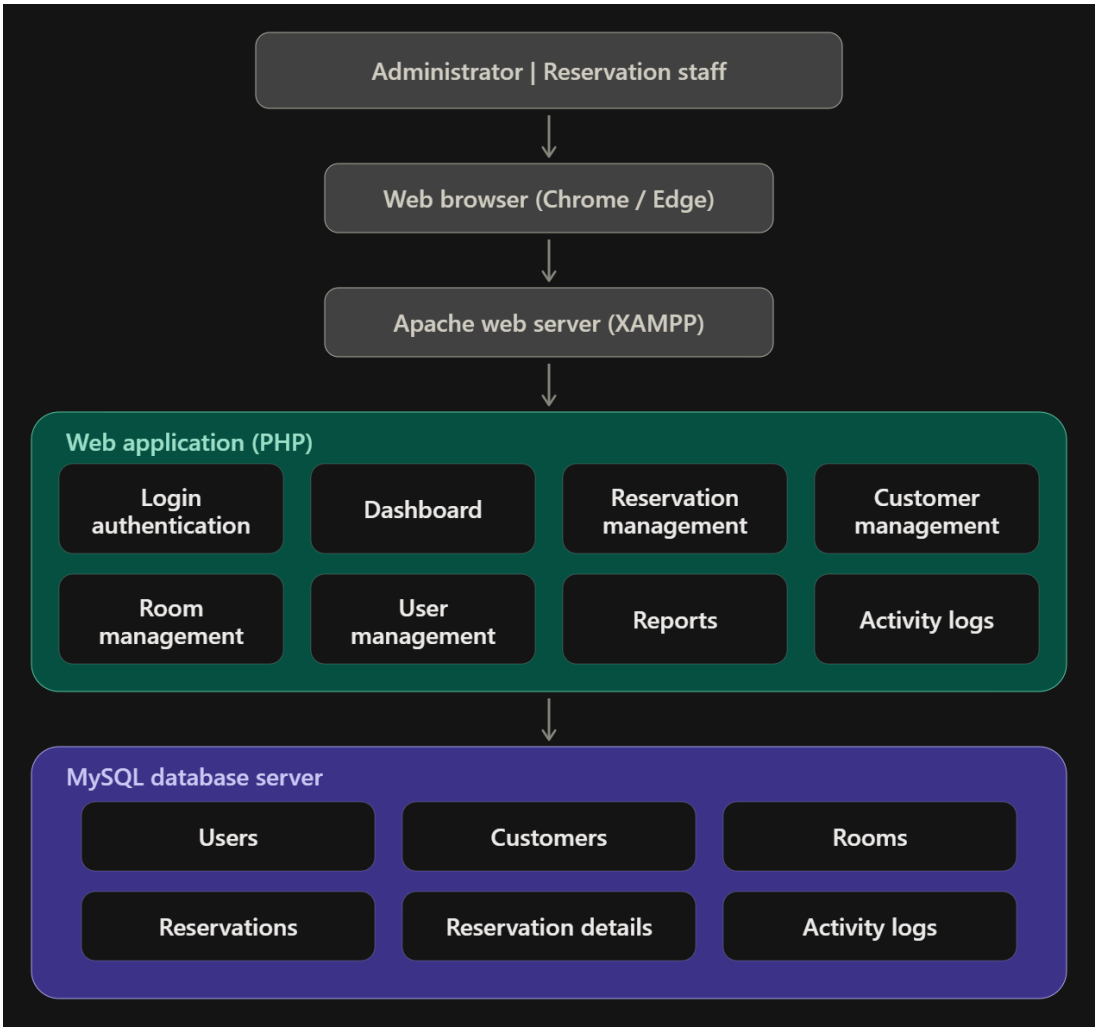


Figure 3.2. System architecture of the web-based reservation system for Coravergel Inland Resort.

3.3 Research Design

This study uses a descriptive-developmental research design, which suits studies that build and evaluate a technological product to solve a practical problem. The design has been applied to recent information system projects in the Philippines. One example is an information management system for a state university that combined a descriptive-developmental design with the Rapid Application Development (RAD) model. Data were gathered through interviews and an ISO/IEC 25010-based questionnaire, and the system was evaluated on functional suitability, reliability, usability, efficiency, maintainability, portability, compatibility, and security [14]. Another example is a web-based school information system developed with an iterative waterfall approach and rated by staff,

learners, parents, and IT specialists against ISO 25010 quality standards [15]. Both match the aim of this study, which is to build and evaluate a secure, centralized, web-based reservation system for Coravergel Inland Resort.

The study follows six phases, adapted from the structured stages used in comparable hotel reservation projects [16].

1. **Analysis.** The researchers examine the resort's existing manual workflow and identify its inefficiencies, limitations, and security weaknesses. Requirements are gathered from staff and management through observation, interviews, and review of related literature, the same data collection methods used in a recent hotel reservation study [16].
2. **Design.** The researchers plan the system architecture, database structure, user interface layouts, and security mechanisms, including authentication, role-based access control, and activity logging. Privacy and security are planned at this stage and not added afterwards. This follows a review of Philippine National Privacy Commission issuances that recommends embedding privacy into the design lifecycle [18]. It is also consistent with research showing that hotels need risk assessment together with technical and organizational safeguards [19].
3. **Development.** The researchers build the system with web technologies and a centralized database. Key features include user authentication, access control, reservation management, real-time room availability, and audit logs.
4. **Testing.** The researchers test the system for functionality, usability, reliability, and security. Black box testing of login, data management, and reservation processes has been used in a comparable hotel system [3]. Security testing in this study is limited to authentication, access control, password protection, and activity logging, as stated in the delimitation.
5. **Implementation.** The system is deployed in the resort's operating environment, where selected staff and administrators use it in actual reservation work.
6. **Evaluation.** The system is evaluated by end users and IT experts against the ISO/IEC 25010 software quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability [14], [15].

The process is iterative. Feedback from users is collected during development and used to refine the system before final deployment. A recent Philippine reservation system for a resort used Agile Scrum with 50 participants, 18 IT professionals and 32 end users, to design and evaluate the system step by step [17]. Iteration lowers the risk of system failure or user dissatisfaction. It also keeps the design user-centered, because users shape it throughout.

This design evaluates the system on its practical usefulness as well as its technical performance. It checks whether the system reduces booking errors, improves record accuracy, speeds up transactions, and protects guest data in actual resort operations.

3.4 System Development Methodology

3.4.1 Chosen SDLC Model: Rapid Application Development (RAD)

This study adopts the Rapid Application Development (RAD) model to develop the web-based reservation system with security features for Coravergel Inland Resort. RAD builds working prototypes quickly, involves users throughout development, and refines the system through repeated cycles of design and testing. A recent web-based reservation system for a sports facility, which replaced a manual process, reported that RAD supported faster and more flexible development and reduced schedule conflicts [20]. A web-based registration system built with RAD likewise began with observations and interviews to identify the constraints of the manual process, and was then tested at each stage [21].

RAD suits this project for three reasons. First, the requirements of resort staff may change as they see the system working, and RAD allows earlier phases to be revisited. A sequential model such as Waterfall requires each stage to be completed before the next begins [22], which makes late changes costly. Second, RAD involves resort administrators and staff in prototype review, so the final system reflects actual workflows. Third, it allows security to be built in and tested during every cycle. Recent Philippine guidance on data privacy recommends embedding privacy and security into the design lifecycle and not treating them as an afterthought [23]. Research on hotel cybersecurity also stresses risk assessment and technical and organizational safeguards [24]. Testing authentication, access control, and logging in each iteration follows both recommendations.

3.4.2 Phases of the RAD Model

TABLE I
RAD PHASES AND OUTPUTS

Phase	Main Activities	Output
1. Requirements Planning	Interviews, observation, document review, security requirements	Requirements specification

2. User Design	Prototyping, user feedback, DFD, ERD, flowcharts, use cases	Approved system design
3. Construction	Module development, integration, security features, continuous testing	Working system
4. Cutover	Unit, integration, system, and user acceptance testing; deployment; staff training; maintenance	Deployed and evaluated system

Phase 1: Requirements Planning. The researchers work with the resort owner, management, and staff to understand how reservations are currently handled. They use interviews, observations, and review of existing records, the same approach used in earlier web-based system studies [21], [25]. They document problems such as duplicate bookings, misplaced records, slow retrieval, and the absence of security controls. The functional requirements are reservation and booking management, real-time room availability, reservation approval and cancellation, user authentication, reporting, and administrative tools. The non-functional requirements are security and data protection, accuracy, a user-friendly interface, fast response, browser accessibility, and database integrity with backup. The scope, constraints, and security requirements (confidentiality, integrity, and availability) are also defined. The output is a requirements specification.

Phase 2: User Design. The researchers convert the requirements into prototypes. These include the login screen, reservation forms, a room availability dashboard, an administrative panel, the database structure, and navigation flows. Staff and management review each prototype and suggest changes, and the prototype is revised over several iterations until it is approved. Design documents are produced alongside it: Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), system flowcharts, and use case diagrams. A comparable reservation system used Figma for interface design and UML for modeling [20]. The output is an approved design.

Phase 3: Construction. The approved design is built with HTML, CSS, and JavaScript for the interface, Python for the backend, and MySQL for the centralized database. The main modules are:

1. **User authentication,** covering registration of staff accounts, login, password validation, and role-based access control.
2. **Reservation management,** covering creation, update, cancellation, and status tracking by staff.
3. **Real-time availability monitoring,** which updates room status immediately when a reservation is made, changed, or cancelled.
4. **Database management,** which stores guest records, reservations, room data, and activity logs.

5. **Security features,** including password hashing, session management, input validation, activity logging, and data backup and recovery.

Each module is tested as it is built, and errors are corrected before integration.

Phase 4: Cutover. Testing is conducted in four levels:

- **Unit testing** checks individual components, such as login, form validation, database insertion and retrieval, and availability updates. A comparable RAD project used PHPUnit for this level [21].
- **Integration testing** verifies that the interface, backend, and database work together. For example, a new reservation must be stored correctly, update room availability, and appear immediately on the dashboard. Cancellations and edits must update all related records without inconsistency.
- **System testing** evaluates the complete workflow: login, booking, availability monitoring, modification and cancellation, guest records, reports, user administration, and access control. Black box testing has been used for this purpose in comparable studies [21].
- **User acceptance testing (UAT)** has selected resort staff and administrators to test the system against real reservation scenarios.

Security testing is limited to authentication, access control, password protection, and activity logging, as stated in the delimitation. After testing, the system is installed on the designated server, the database is configured, user accounts are created, and existing room data and reservation records are migrated. Staff then receive orientation on secure login, reservation handling, room updates, reports, and account management, and a user manual is provided.

3.4.3 Evaluation and Continuous Improvement

The deployed system is evaluated against the ISO/IEC 25010 software quality characteristics, which comparable system development studies have used for end-user evaluation, including security [25]. Feedback is collected from administrators and staff, and recent Philippine work has used iterative cycles with both IT professionals and end users as evaluators [26]. After deployment, the researchers carry out maintenance activities: bug fixes, monitoring of database performance, updates to authentication and password policies, and performance tuning. Online payment, customer self-service, and automated notifications are excluded from this study and are recommended only as future work.

3.5 Data Collection and Traffic Injection

This section explains how data were gathered for the study and how controlled test traffic was injected into the developed reservation system to evaluate its performance and security controls. It has two parts.

Data collection covers the information gathered about the resort's manual reservation process and the information produced by the developed system.

Traffic injection means the deliberate sending of simulated requests to the developed system, inside an isolated test environment, to observe how it behaves. The simulated requests include normal user workflows under load and test requests that check whether validation, authentication, role-based access control (RBAC), and activity logging work as designed. Traffic injection here is a defensive testing technique. It is not used against the live resort system or against any system the researchers do not own.

The approach is limited to the controls described in the study's scope: authentication, access control, session management, activity logging, and input validation. It does not include comprehensive penetration testing, which the study lists as a delimitation.

3.5.1 Data Collection

Types of Data

The study uses two groups of data.

Group	Source	Purpose
Problem and requirements data	Interviews, surveys, observation, document analysis	Describe the manual process and its weaknesses, and define system requirements
System-generated data	Test results, performance records, scanner reports, database records, activity logs	Evaluate the developed system's performance, security, and reliability

Problem and Requirements Data

These methods were described in Chapter 3 and are summarized here.

1. Interviews. Structured and semi-structured interviews were held with the resort owner, management, and reservation staff about how bookings are recorded, updated, and protected.
2. Surveys. Questionnaires collected measurable feedback on satisfaction, frequency of errors, booking time, and accuracy of availability information.
3. Observation. The researchers watched the daily reservation workflow without interfering, noting how records are written, searched, and corrected.
4. Document analysis. Existing logbooks, booking forms, and spreadsheets were reviewed for incomplete, duplicated, or inconsistent records.

Using several methods allows findings from one source to be checked against the others. This matters for security in particular. In the security testing study by Qadir et al. [20], none of the nine automated tools detected problems in the logging and monitoring category, and the authors noted that this category is usually assessed through interviews or by checking whether attacks were noticed. This supports using interviews and document review alongside technical testing, rather than relying only on tools.

System-Generated Data

Data	Collected from	Used for
Response time, throughput, error rate	Load test result files	Performance efficiency
Scanner alerts and severity levels	Security scan reports	Security evaluation
Pass/fail outcomes of test cases	Test case records	Functional and security validation
Login, reservation, and administrative events	Activity Logs table	Verifying accountability and traceability
Reservation and room status records	Reservations and Rooms tables	Checking data integrity and double-booking prevention
Web server access and error logs	Apache logs	Cross-checking test traffic against recorded events

Data Handling and Privacy

Reservation records contain personal information such as names, contact details, and booking dates. Hotels hold sensitive data of this kind in their reservation databases, and human error is often a major factor in data breaches [21]. A 2023 industry report found that about 31 percent of hospitality organizations had experienced a data breach [22]. For these reasons, the following rules apply:

- Real guest records are never used for traffic injection. A test copy of the database with fictional guest data is used.
- Test accounts are created for each role (administrator and reservation staff) and removed after testing.
- Collected interview and survey data are stored securely, reported in aggregate, and used only for this study, consistent with the data privacy practices discussed in Chapter 1.
- Written permission from the resort owner is obtained before any testing.

3.5.2 Traffic Injection

Purpose

Traffic injection answers four questions:

1. Does the system stay responsive and stable when several users act at once?
2. Does the system prevent two reservations for the same room and dates when requests arrive together?
3. Do authentication, RBAC, session expiry, and input validation block requests they are meant to block?
4. Are the actions generated during testing recorded in the activity log?

Test Environment

The system is deployed on an isolated machine using the same stack described in Chapter 4: Apache and MySQL under XAMPP, with the PHP application. Testing is not performed on the resort's live system. Qadir et al. [1] used the same approach, deploying each application locally on XAMPP and MySQL, and they note that a local setup may not reproduce production factors such as different server settings or heavier traffic. This limitation is carried into this study (see Section 5).

Item	Setting
Server	XAMPP (Apache, MySQL, PHP) on an isolated test machine
Database	Test copy with fictional data and a backup taken before each test run

Accounts	Test administrator and test staff accounts
Network	Local or isolated network only
Tools	Apache JMeter for load traffic [21]; OWASP ZAP for security scanning [20]

Tools

Apache JMeter generates simulated user load against web applications. Alnuhait et al. [23] used JMeter to test two websites and measured responsiveness, throughput, and scalability. They observed that response time grew as test duration and load increased, and that a configuration with a 10-second ramp time and five loop iterations produced significant performance degradation. These parameters, ramp time and loop count, are therefore recorded for every run in this study.

OWASP ZAP is a dynamic application security testing (DAST) tool that examines a running application. In the comparison by Qadir et al. [20], using DAST alone was the preferred option for four OWASP Top 10:2021 categories, and ZAP was the most effective DAST tool, especially for broken access control, insecure design, and security misconfiguration. The same study also found that no tool detected issues in every category. A separate master's thesis comparing ZAP, Arachni, and Wapiti under black-box conditions found ZAP's detection score lower than the other two tools [24]. ZAP is therefore used as one source of evidence and is supported by manual test cases.

Traffic Types

#	Traffic type	Description	What it checks
T1	Normal workflow traffic	Simulated staff logging in, checking availability, creating, editing, and cancelling reservations, and opening reports	Functional correctness and baseline performance
T2	Concurrent load	Several simulated users performing T1 actions at the same time, with load increased in stages	Responsiveness, throughput, stability
T3	Concurrent booking of the same room	Multiple simulated requests to reserve the same room and dates at nearly the same moment	Double-booking prevention and data integrity

T4	Invalid input	Empty fields, wrong date formats, check-out earlier than check-in, invalid contact numbers and emails	Input validation
T5	Unauthorized role requests	A staff-role session requesting administrator-only functions, and requests with no login	Authentication and RBAC
T6	Session tests	Requests made after logout or after the idle timeout	Session management
T7	Automated security scan	ZAP passive and active scan of the test instance	Common web vulnerabilities and misconfiguration

T5 corresponds to broken access control, which is ranked first in the OWASP Top 10:2025, the first update to the list since 2021 [25], [26]. The OWASP leads advise denying access by default except for public resources, and they report that broken access control was found in about 3.73 percent of the applications tested [26]. The 2025 list also includes categories for authentication failures and for logging and alerting failures [25], which correspond to T5, T6, and the log verification in Section 3.6.

Procedure

1. Back up the test database and confirm the application runs normally.
2. Prepare fictional guest and room data and the test accounts.
3. Record the baseline: server settings, tool versions, and number of existing records.
4. Run T1 and T2 with JMeter, starting with a small number of simulated users and increasing in stages. Record the ramp time, loop count, and number of users for each run.
5. Run T3 to T6 as defined test cases, each with an expected result written beforehand (for example, "second booking for the same room and dates is rejected").
6. Run the ZAP scan (T7) on the test instance only.
7. Collect JMeter result files, ZAP reports, database state, Apache logs, and the Activity Logs table.
8. Restore the database from backup before the next run so that results are comparable.
9. Repeat each run at least three times and record all results, including failures.

Verifying Logging

Automated tools did not detect weaknesses in the logging category in the study by Qadir et al. [20], so logging is verified manually. After the traffic runs, the researchers compare the events they generated with what appears in the Activity Logs table: logins, failed logins, reservation creation, changes, cancellations, and attempts to reach restricted functions. A test passes when each generated event has a log entry showing the user, action, date and time, and IP address where available.

3.5.3 Data Analysis

1. Performance. Average, highest, and percentile response times, throughput, and error rate are compared across user levels to find the point where performance starts to fall, following the responsiveness, throughput, and scalability measures used in [23].
2. Functional and security tests. The pass rate is computed for each traffic type. Every failed test is described, fixed, and retested.
3. Scanner findings. Findings are grouped by severity and mapped to OWASP Top 10 categories, as done in [20]. Each finding is reviewed manually to separate genuine issues from false alarms.
4. Quality characteristics. Results are summarized against ISO/IEC 25010:2023, which defines nine product quality characteristics, including performance efficiency, security, reliability, and interaction capability (the renamed usability characteristic) [27]:

Evidence	ISO/IEC 25010:2023 characteristic
Load test results	Performance efficiency
Concurrent booking and repeated runs	Reliability
RBAC, session, validation, and log tests	Security
Survey and user acceptance feedback	Interaction capability

3.5.4 Limitations

- Testing is done in a local test environment and may not reflect the conditions of a live deployment [20].
- Automated scanners do not find every kind of weakness, and different tools can give different results [20], [24].
- Traffic injection is limited to the implemented controls and is not a full penetration test, in line with the study's delimitation.

- Results describe this system, this configuration, and this hardware, and should not be generalized to other resorts without further testing.

3.6 Hardware and Software Environment

The system uses a three-tier client-server design. A web browser serves as the client, an Apache and PHP application handles business rules and security controls, and a MySQL or MariaDB database stores all records. The researchers used three environments:

- **Development environment:** where the modules are built and unit-tested.
- **Test environment:** an isolated machine with fictional data for load, validation, access-control, and scanner testing. Qadir et al. [27] also deployed their test applications locally on XAMPP and MySQL, and noted that a local setup may not reproduce production traffic or server settings. That limitation applies here too.
- **Deployment environment:** the resort's host computer or server, running on the local network or an internet-accessible server (see Section 3.2).

Security requirements were considered when choosing every component, in line with NIST's recommendation to integrate secure development practices into the software life cycle [28] and the guidance on embedding privacy and security into system design for Philippine data privacy compliance [29].

3.5 Detection Algorithm and Sliding-Window Execution

[9][10]Detection is realized by combining the existing activity-logging module with a lightweight sliding-window anomaly detector. Every critical system event (successful/failed login, reservation create/update/delete, role-based access denial, and input-validation failure) is written to the Activity Logs table.

A fixed-size sliding window of length WWW (default W=50W = 50W=50 events or a 5-minute time window, whichever is reached first) is maintained over the most recent log entries, both per user session and globally. Within each window the following features are extracted:

- Number of failed authentication attempts
- Number of input-validation failures
- Frequency of privilege-related denials
- Rate of actions compared with the user's historical baseline

- Entropy of requested endpoints and parameters

[11][12]If any feature exceeds a pre-calibrated threshold (or a statistical deviation threshold derived from normal traffic), the current window is classified as anomalous and an alert is generated for the administrator. The window then advances by a step size SSS (typically S=W/2S = W/2S=W/2), enabling continuous monitoring with temporal context.

This design is intentionally lightweight so that it can execute on the same Apache/PHP environment used by the reservation system without requiring a separate heavy intrusion-detection engine. It complements the preventive controls already implemented (password hashing, role-based access control, and input validation).

3.7 Performance and Evaluation Metrics

The performance of the sliding-window detection mechanism was evaluated using standard classification metrics on the labeled dataset:

- Detection Rate / True Positive Rate (TPR)
- False Positive Rate (FPR)
- Accuracy
- Precision, Recall, and F1-score
- Average detection latency (time from the occurrence of an anomalous event to the generation of an alert)

[13]Both offline (batch analysis of collected logs) and near-real-time (continuous sliding-window execution during live test sessions) modes were measured. In addition, resource overhead (CPU and memory consumption on the XAMPP test environment) was recorded to verify that the detector remains practical for a small-to-medium resort deployment.

The evaluation confirmed that the combination of preventive security controls and the lightweight sliding-window detector provides effective protection against the most common threats targeting web-based reservation systems, while introducing only modest computational overhead.

REFERENCE

[1] S. Qadir *et al.*, "Comparative evaluation of approaches & tools for effective security testing of Web applications," *PeerJ Comput. Sci.*, vol. 11, p. e2821, 2025, doi: 10.7717/peerj-cs.2821.

[2] E. Hertzfeld, "Staff issues drive need for cybersecurity," *Hotel Management*, Jun. 2023. [Online]. Available: https://hotelmanagement.mydigitalpublication.com/articles/security

[3] "Report: 31% of hospitality organizations have had a data breach," *Hotel Management*, 2023. [Online]. Available: https://www.hotelmanagement.net/tech/report-31-hospitality-organizations-have-had-data-breach

[4] H. Alnuhait, W. Alzyadat, A. Althunibat, H. Kahtan, B. Zaqaibeh, and H. A. Al-Khawaja, "Web application performance assessment: A study of responsiveness, throughput, and scalability," *Int. J. Adv. Appl. Sci.*, vol. 11, no. 9, pp. 214–226, Sep. 2024, doi: 10.21833/ijaas.2024.09.023.

[5] C. Sharma, "A comparative analysis of OWASP ZAP, Arachni and Wapiti against the Reinforced WAVSEP benchmark," M.S. thesis, School of Computing, National College of Ireland, Dublin, Ireland, 2025. [Online]. Available: https://norma.ncirl.ie/9832/

[6] OWASP Foundation, "OWASP Top 10:2025," 2025. [Online]. Available: https://owasp.org/Top10/

[7] "OWASP Top 10: Broken access control still tops app security list," *The Register*, Nov. 11, 2025. [Online]. Available: https://www.theregister.com/2025/11/11/new_owasp_top_ten_broken/

[8] ISO/IEC, *Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model*, ISO/IEC 25010:2023, Nov. 2023.

[9] S. Meganathan, A. Sumathi, S. Sheik Mohideen Shah, and R. Rajakumar, “Real-Time Intrusion Detection Using Adaptive Sliding Windows for Concept Drift,” *Int. J. Electron. Commun. Eng.*, vol. 12, no. 8, 2025.

[10] “Hybrid network intrusion detection system based on sliding window and information entropy in imbalanced dataset,” *Appl. Intell.*, 2025.

[11] “User Behavior Threat Detection Based on Adaptive Sliding Window GAN,” *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 2, pp. 2493–2503, Apr. 2024, doi: 10.1109/TNSM.2024.3355698.

[12] "A Novel Real-Time Web Based Intrusion Detection System (IDS)," in *Proc. 4th Int. Multidisciplinary Information Technology and Engineering Conf. (IMITEC)*, 2024, doi: 10.1109/IMITEC60221.2024.10850945.

[13] "Deep Learning for Web Application Security: A Comprehensive Survey on Intrusion Detection Systems," in *Proc. 2nd Int. Conf. Intelligent Systems for Cybersecurity (ISCS)*, 2025.

[14] A. R. N. Malaya, E. A. Munar, F. P. Cuison, and E. G. Dacanay, "Information management system for research of Don Mariano Marcos Memorial State University–South La Union Campus," Indones. J. Electr. Eng. Comput. Sci., vol. 28, no. 3, pp. 1668–1675, 2022, doi: 10.11591/ijeecs.v28.i3.pp1668-1675.

[15] K. Caratiquit, "Web-based school information and publication system: A developmental study," Global Educ. Soc. Sci. J., vol. 1, no. 3, pp. 45–55, 2021.

[16] E. M. Benolon, Y. P. K. Kelen, and S. S. Manek, "Design and development of Ariesta Hotel reservation information system based website using waterfall method," J-ENSITEC (J. Eng. Sustain. Technol.), vol. 12, no. 2, pp. 10497–10507, Jun. 2026, doi: 10.31949/j-ensitec.v12i02.18002.

[17] E. Balintag, K. J. H. Concepcion, P. J. Dagman, J. A. Ducos, D. Fernandez, and E. Villanueva, "SORMS: Steffi's Online Reservation Management System," Psychol. Educ.: A Multidiscip. J., vol. 36, no. 7, 2025. [Online]. Available: https://ejournals.ph/article.php?id=32526

[18] D. Cheng, "Managing data privacy compliance and technology rollout in the Philippines post COVID-19," in Proc. De La Salle Univ. Res. Congress, 2023. [Online]. Available: https://www.dlsu.edu.ph/wp-content/uploads/pdf/conferences/research-congress-proceedings/2023/HCT-01.pdf

[19] K. Pipyros and S. Liasidou, "A new cybersecurity risk assessment framework for the hospitality industry: Techniques and methods for enhanced data protection and threat mitigation," Worldwide Hospitality Tourism Themes, vol. 17, no. 1, pp. 48–61, Mar. 2025, doi: 10.1108/WHATT-12-2024-0296.

[20] A. Handika Mulyana, R. Dwi Cahya Supriatman, and T. Rohayati, "Sistem reservasi lapangan olahraga berbasis web menggunakan metode Rapid Application Development (RAD) (studi kasus: Kawali Sport Center)," J. Mahasiswa Sist. Inf. Galuh, vol. 2, no. 2, pp. 33–48, 2026, doi: 10.25157/jmsig.v2i2.5482.

[21] I. Riadi, A. Yudhana, and A. Elvina, "Analysis impact of Rapid Application Development method on development cycle and user satisfaction: A case study on web-based registration service," Sci. J. Informatics, vol. 11, no. 1, pp. 81–94, 2024, doi: 10.15294/sji.v11i1.49590.

[22] E. M. Benolon, Y. P. K. Kelen, and S. S. Manek, "Design and development of Ariesta Hotel reservation information system based website using waterfall method," J-ENSITEC (J. Eng. Sustain. Technol.), vol. 12, no. 2, pp. 10497–10507, Jun. 2026, doi: 10.31949/j-ensitec.v12i02.18002.

[23] D. Cheng, "Managing data privacy compliance and technology rollout in the Philippines post COVID-19," in Proc. De La Salle Univ. Res. Congress, 2023. [Online]. Available: https://www.dlsu.edu.ph/wp-content/uploads/pdf/conferences/research-congress-proceedings/2023/HCT-01.pdf

[24] K. Pipyros and S. Liasidou, "A new cybersecurity risk assessment framework for the hospitality industry: Techniques and methods for enhanced data protection and threat mitigation," Worldwide Hospitality Tourism Themes, vol. 17, no. 1, pp. 48–61, Mar. 2025, doi: 10.1108/WHATT-12-2024-0296.

[25] A. R. N. Malaya, E. A. Munar, F. P. Cuison, and E. G. Dacanay, "Information management system for research of Don Mariano Marcos Memorial State University–South La Union Campus," Indones. J. Electr. Eng. Comput. Sci., vol. 28, no. 3, pp. 1668–1675, 2022, doi: 10.11591/ijeecs.v28.i3.pp1668-1675.

[26] E. Balintag, K. J. H. Concepcion, P. J. Dagman, J. A. Ducos, D. Fernandez, and E. Villanueva, "SORMS: Steffi's Online Reservation Management System," Psychol. Educ.: A Multidiscip. J., vol. 36, no. 7, 2025. [Online]. Available: https://ejournals.ph/article.php?id=32526

[27] S. Qadir et al., "Comparative evaluation of approaches & tools for effective security testing of Web applications," *PeerJ Comput. Sci.*, vol. 11, p. e2821, 2025, doi: 10.7717/peerj-cs.2821.

[28] M. Souppaya, K. Scarfone, and D. Dodson, "Secure Software Development Framework (SSDF) version 1.1: Recommendations for mitigating the risk of software vulnerabilities," Nat. Inst. Standards Technol., Gaithersburg, MD, USA, NIST SP 800-218, Feb. 2022, doi: 10.6028/NIST.SP.800-218.

[29] D. Cheng, "Managing data privacy compliance and technology rollout in the Philippines post COVID-19," in *Proc. De La Salle Univ. Res. Congress*, 2023. [Online]. Available: https://www.dlsu.edu.ph/wp-content/uploads/pdf/conferences/research-congress-proceedings/2023/HCT-01.pdf

CHAPTER IV

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the implementation and evaluation framework for the Secure Web-Based Reservation Management System developed for Coravergel Inland Resort. The system is intended to address the problems identified in the existing manual process, including fragmented records, slow retrieval of booking information, inaccurate room-status updates, duplicate reservations, and limited protection of guest data. The proposed application centralizes reservation, room, guest, user-account, and activity-log information in a web-based environment.

The chapter describes the system modules, architecture, implementation technologies, database responsibilities, security controls, and test cases used to assess whether the application meets its requirements. The discussion follows the scope established in the research paper: the platform is primarily for resort administrators and authorized staff, and it does not include public self-service booking, online payment gateways, external travel-platform integration, or a dedicated mobile application.

Security and quality checks are framed using recent guidance. NIST recommends integrating secure development practices throughout the software development life cycle [1]. The NIST Cybersecurity Framework 2.0 provides a current structure for identifying, protecting, detecting, responding to, and recovering from cybersecurity risks [2]. The OWASP Top 10:2021 and Application Security Verification Standard (ASVS) 5.0.0 provide relevant checkpoints for access control, authentication, input handling, logging, and application security verification [3], [4].

4.2 Overview of the Developed System

The developed system is a secure web-based reservation management application designed to replace the manual reservation procedures currently practiced at Coravergel Inland Resort. The application centralizes reservation management by providing authorized personnel with a single platform for processing guest reservations, monitoring room availability, managing customer information, generating reports, and maintaining reservation records. Through automation, the system minimizes the risks associated with manual documentation while improving operational efficiency, reservation accuracy, and information security.

Unlike the previous manual system, where reservation information is recorded using paper documents and spreadsheets, the developed application stores all operational data within a centralized relational database. This allows authorized users to access updated reservation information in real time, significantly reducing delays in information retrieval, duplicate reservations, inconsistent records, and communication errors among resort personnel.

The system provides different levels of access according to user responsibilities. System Administrators possess complete administrative privileges, allowing them to manage user accounts, monitor reservation activities, generate reports, configure system settings, and oversee security controls. Reservation Staff and Front Desk Personnel are granted access only to functions necessary for their assigned operational responsibilities, thereby implementing the principle of least privilege and reducing unauthorized system access.

To strengthen cybersecurity, the system incorporates several security mechanisms including secure user authentication, encrypted password storage using hashing algorithms, role-based access control, activity logging, session management, and automated database backup support. These security features help protect confidential customer information while maintaining accountability among authorized users.

4.2.1 Main System Modules

Module	**Purpose**	**Expected output**
Authentication and user management	Verifies user credentials and manages authorized accounts and roles.	Authenticated session and role-appropriate access.
Guest information management	Creates, retrieves, and updates guest records needed for reservations.	Consistent guest profiles linked to reservations.
Room management	Maintains room details and room-status information.	Current room records and availability status.
Reservation management	Creates, views, updates, and manages reservation records.	Organized reservation history and current booking status.

Reports and summaries	Provides basic reservation, occupancy, and activity summaries for management.	Readable operational reports based on stored records.
Activity logging	Records relevant user actions, such as reservation changes and administrative activity.	Traceable event history for accountability and troubleshooting.
Database backup and maintenance	Supports preservation and recovery of important records according to the deployment procedure.	Recoverable data when backups are configured and verified.

The listed modules are consistent with the functions described in the research document. The final implementation should be checked against the actual application screens and source code so that any feature not yet implemented is identified as pending rather than described as completed.

The system provides different levels of access according to user responsibilities. System Administrators possess complete administrative privileges, allowing them to manage user accounts, monitor reservation activities, generate reports, configure system settings, and oversee security controls. Reservation Staff and Front Desk Personnel are granted access only to functions necessary for their assigned operational responsibilities, thereby implementing the principle of least privilege and reducing unauthorized system access.

4.4 System Development Methodology

The system was developed using a phased software development life cycle consisting of requirements gathering, system planning, software development, testing, and maintenance. Security activities were included in every phase rather than added at the end, following the practice of integrating secure development tasks into the life cycle recommended by NIST [1].

4.4.1 Requirements Gathering

Requirements were gathered by reviewing the resort's existing booking procedures, the paper forms and spreadsheets in use, and the problems stated in Section 4.1. Functional requirements include recording guests and

reservations, preventing overlapping bookings, tracking room status, assigning roles, logging user actions, and producing summaries. Non-functional requirements include confidentiality of guest data, accountability of users, availability of records, ease of use by non-technical staff, and recoverability from failure.

The first phase involved collecting information regarding the current reservation management practices of Coravergel Inland Resort. The researchers conducted interviews with the resort owner and reservation personnel to understand the existing workflow, operational challenges, and security concerns associated with the manual reservation system.

Direct observation of reservation activities was also performed to identify the sequence of reservation processing, room allocation procedures, customer information recording, and report preparation. Existing reservation forms, guest records, and manual logbooks were reviewed to determine the information required by the proposed system.

Based on the collected information, the following functional requirements were identified:

- User authentication
- User account management
- Customer management
- Reservation management
- Room management
- Reservation status monitoring
- Report generation
- Activity logging
- Database backup

The non-functional requirements identified include:

- Data confidentiality
- System availability

· Fast response time

· Easy-to-use interface

· Secure data storage

· Reliable database performance

· Scalability

· Maintainability

The collected requirements served as the foundation for designing and developing the proposed reservation management system.

4.4.2 System Planning

Planning translated the requirements into a scope, module list (Table 4.1), user roles, and a schedule. The scope boundaries stated in Section 4.1 were fixed at this stage: no public self-service booking, no online payment gateway, no external travel-platform integration, and no dedicated mobile application. The vulnerabilities in Table 4.2 were used to prioritize which controls were designed first, giving priority to the High-rated items.

After gathering the system requirements, the researchers prepared a comprehensive development plan that defined the project's scope, objectives, deliverables, timeline, and required resources.

During this phase, system modules were identified and organized according to the operational requirements of Coravergel Inland Resort. The researchers also determined the technologies required for development, including the programming language, database management system, development environment, and supporting software.

Potential risks associated with the project were also identified. These included possible data migration issues, unauthorized access to sensitive information, system implementation delays, hardware limitations, and user adaptation challenges. Appropriate mitigation strategies were incorporated into the development plan to minimize these risks.

4.4.3 System Design

The system design phase translated the identified requirements into a complete software blueprint.

The researchers designed the overall system architecture, database structure, navigation flow, user interfaces, and security framework. Flowcharts, Entity Relationship Diagrams (ERD), Data Flow Diagrams (DFD), and Use Case Diagrams were utilized to visualize system processes and database relationships.

Special consideration was given to cybersecurity during the design process. Security controls were integrated into the system architecture to support:

- Confidentiality
- Integrity
- Availability

User authentication, role-based access control, password hashing, session management, audit logging, and backup mechanisms were incorporated into the design to strengthen information security.

The graphical user interface was designed following user-centered design principles to ensure that authorized personnel could easily navigate the application with minimal technical training.

4.4.4 Software Development

Development covered database design, interface design, and coding of each module. Security controls were built into the code: passwords are stored as salted hashes rather than plain text, every page checks the user's session and role, user input is validated and passed to the database through parameterized queries, and significant actions are written to the activity log. Password handling follows the direction of current digital identity guidance, which emphasizes password length and screening over arbitrary composition rules [5].

Following the approved system design, the researchers proceeded with software development.

The system was developed as a web-based application using modern web development technologies. Front-end development focused on creating responsive and user-friendly interfaces that enable authorized users to perform reservation-related tasks efficiently. Back-end development implemented the application's business logic, reservation processing functions, database transactions, report generation, and security mechanisms.

During development, emphasis was placed on secure coding practices to reduce common web application vulnerabilities such as:

- SQL Injection
- Cross-Site Scripting (XSS)
- Unauthorized Access
- Session Hijacking
- Weak Authentication

Password credentials were securely stored using password hashing algorithms rather than plain text storage.

The reservation module automatically validates room availability before confirming reservations, reducing the possibility of duplicate bookings.

Throughout development, each module was tested individually before integration into the complete system.

4.4.5 Testing

Testing was designed to confirm that each vulnerability in Table 4.2 is actually addressed. Functional tests verify that reservations, rooms, and guest records behave correctly. Security tests check login handling, role restrictions, session expiry, input validation, and log entries against the relevant ASVS areas [4]. The test cases are presented in Section 4.9.

System testing was conducted after software development to verfy that the application functioned according to the specified requirements.

The researchers performed several forms of testing including:

Functional Testing

Each system feature was tested individually to verify that it produced the expected output.

Examples include:

· User Login

· Reservation Creation

· Reservation Editing

· Reservation Cancellation

· Room Availability Updates

· Report Generation

· User Account Management

Integration Testing

Integration testing verified that all modules worked together correctly without affecting overall system functionality.

Performance Testing

The researchers evaluated system responsiveness during normal operational workloads to ensure that reservation processing remained efficient.

Security Testing

Security testing focused on verifying the effectiveness of:

· User authentication

· Access control

· Password protection

· Session management

· Activity logging

Unauthorized access attempts were performed using test accounts to verify that restricted functions remained inaccessible to users without sufficient privileges.

Any identified errors were corrected before deployment.

Deployment

Following successful testing, the developed reservation management system was deployed within the operational environment of Coravergel Inland Resort.

The application was installed on the designated web server, while the centralized database was configured to securely store reservation records, customer information, user accounts, room information, and system logs.

Authorized personnel received orientation regarding system operation, reservation processing procedures, user account management, and security responsibilities.

Historical reservation records required for operational continuity were encoded into the database to ensure a smooth transition from manual to computerized reservation management.

4.4.6 Maintenance

Maintenance covers scheduled database backups, periodic restore tests, account review, software updates, and correction of defects found in use. It also covers continued staff training, since a recent hotel case study reported that learning curves and incomplete use of system features limited the benefits of a new property management system [8]. Maintenance activities map to the Recover and Detect outcomes of the NIST Cybersecurity Framework [2].

System maintenance is essential to ensure the continuous operation, reliability, and security of the developed reservation management system.

Routine maintenance activities include:

· Software updates

· Database optimization

· User account management

· Password updates

· Security monitoring

· Database backup

· Bug fixing

· Performance optimization

The researchers recommend periodic system evaluation to identify additional functionality that may improve operational efficiency.

Future maintenance may include the integration of online payment systems, SMS notifications, mobile application support, cloud deployment, and multi-factor authentication to further enhance the system's capabilities.

4.5 System Architecture

The system follows a three-tier web architecture: a browser interface for users, a web application that holds the business rules and security controls, and a relational database that stores all records. Every request from a user passes through the web application, so no user reads or writes the database directly. This single path is what lets the system enforce login, role checks, validation, and logging in one place, which the manual process could not do.

4.6 Hardware and Software Requirements

The successful implementation of the proposed Secure Web-Based Reservation Management System requires appropriate hardware and software resources to ensure stable system performance, efficient reservation processing, and secure management of customer information. The researchers identified the minimum and recommended specifications necessary for system development, deployment, and daily operation.

The hardware requirements focus on providing sufficient computing power to support web application processing, database management, and multi-user access. Likewise, the software requirements include the operating

system, web server, programming language, database management system, and supporting development tools used throughout the development of the application.

The following subsections present the hardware and software requirements of the developed system.

4.6.1 Hardware Requirements

The hardware requirements specify the minimum and recommended computer specifications needed to operate the reservation management system efficiently. The proposed system is designed to operate on standard desktop or laptop computers commonly used in small and medium-sized business establishments such as Coravergel Inland Resort.

Table 4.1

Minimum Hardware Requirements

Component	**Minimum Specification**
Processor	Intel Core i3 (8th Generation) or AMD Ryzen 3
Memory	(RAM) 8 GB DDR4
Storage	256 GB Solid State Drive (SSD)
Monitor	1366 × 768 Resolution
Keyboard	Standard USB Keyboard
Mouse	Optical USB Mouse
Network	Broadband Internet Connection (10 Mbps or higher)
Printer	Inkjet/Laser Printer (Optional for reports)

The minimum hardware specifications are sufficient for processing reservations, maintaining customer records, generating reports, and supporting simultaneous access by authorized personnel under normal operational

conditions. For improved performance and future scalability, the researchers recommend the following hardware specifications.

Table 4.2

Recommended Hardware Requirements

Component	**Minimum Specification**
Processor	Intel Core i5 (10th Generation) or AMD Ryzen 5
Memory	(RAM) 16 GB DDR4
Storage	512 GB SSD
Monitor	Full HD (1920 × 1080)
Keyboard	Mechanical or Standard USB Keyboard
Mouse	Optical Mouse
Network	Fiber Internet Connection (50 Mbps or higher)
Backup	Device External Hard Drive or Cloud Storage
UPS	650VA or Higher Uninterruptible Power Supply

The recommended specifications provide better processing speed, faster database transactions, improved multitasking capability, and enhanced reliability during peak reservation periods.

The inclusion of an Uninterruptible Power Supply (UPS) helps prevent data corruption caused by sudden power interruptions, while external backup storage provides additional protection for critical reservation data.

4.6.2 Software Requirements

The software requirements identify the development tools, operating systems, web technologies, database management software, and application frameworks utilized during the implementation of the reservation management system.

The researchers selected software technologies that are stable, widely supported, cost-effective, and suitable for web-based application development.

Table 4.3

Software	Purpose
Windows 10 / Windows 11	Operating System
XAMPP	Local Web Server Environment
Apache Web Server	Web Hosting Service
PHP	Server-Side Programming Language
HTML5	Web Structure
CSS3	User Interface Design
Bootstrap 5	Responsive User Interface Framework
JavaScript	Client-Side Interactivity
MySQL	Database Management System
phpMyAdmin	Database Administration
Visual Studio Code	Source Code Editor
Google	Chrome / Microsoft Edge Web Browser
Git (Optional)	Version Control

4.7 Database Design

The database design serves as the foundation of the proposed Secure Web-Based Reservation Management System. It provides a structured method for storing, organizing, retrieving, and managing reservation-related information while maintaining data consistency, integrity, and security.

A relational database model was adopted to establish relationships among different entities involved in the reservation process. The database was designed using normalization principles to eliminate redundant data, improve storage efficiency, and maintain referential integrity between related tables.

Each table represents a specific entity within the system and is connected through primary and foreign keys to ensure accurate data relationships. The database supports real-time reservation processing, customer information management, room availability monitoring, report generation, and activity logging.

The following sections describe the purpose and contents of each database table.

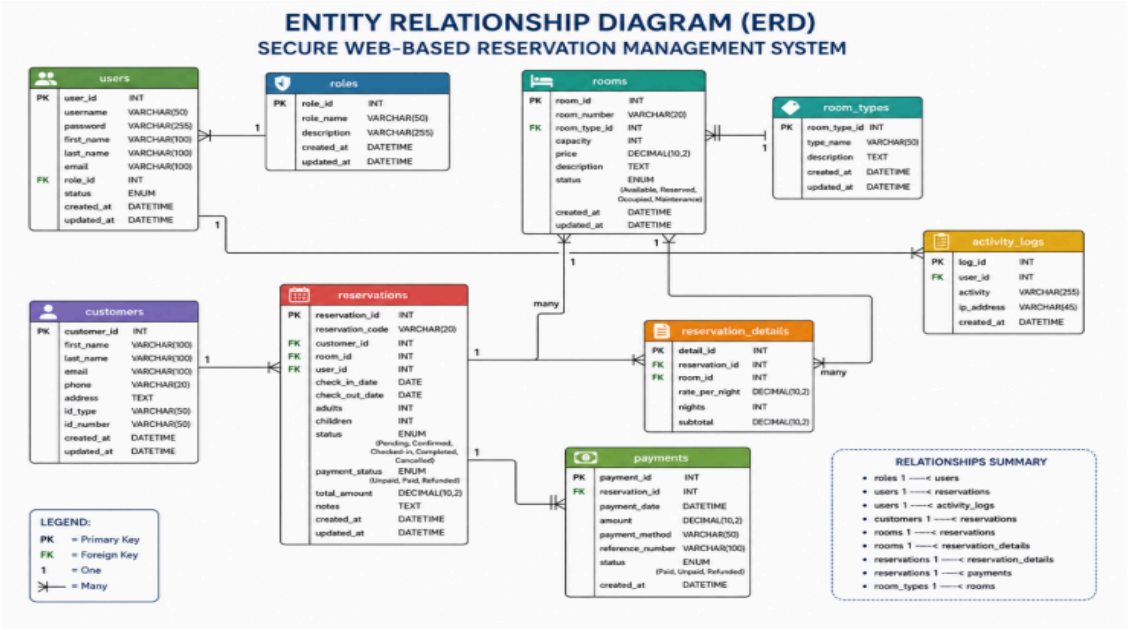


Figure 4.2 illustrates the Entity Relationship Diagram (ERD) of the proposed database .

4.7.1 Users Table

The Users Table stores the accounts of authorized personnel and their roles.

Field	Data type	Key	Description
user_id	INT, auto-increment	PK	Unique account identifier
full_name	VARCHAR(100)		Name of the staff member
username	VARCHAR(50)	Unique	Login name
password_hash	VARCHAR(255)		Salted hash of the password; the plain password is never stored
role	ENUM		Administrator, Reservation Staff, or Front Desk
status	ENUM		Active or Inactive
last_login	DATETIME		Time of the most recent successful login
created_at	DATETIME		Time the account was created

4.7.2 Customers Table

The Customers table stores personal information about guests who make reservations at Coravergel Inland Resort. This information enables the system to process bookings, maintain customer records, and generate reservation reports.

Each customer is assigned a unique customer identification number to prevent duplicate records and simplify future reservations.

Table 4.5

Field Name	Data Type	Description
customer_id	INT (Primary Key)	Customer identification number
first_name	VARCHAR(50)	Customer first name
last_name	VARCHAR(50)	Customer last name
address	TEXT	Residential address
contact_number	VARCHAR(20)	Mobile number
email	VARCHAR(100)	Email address
valid_id	VARCHAR(100)	Government-issued ID reference
created_at	TIMESTAMP	Record creation date

The information stored in this table supports reservation processing while ensuring customer records remain organized and easily retrievable.

4.7.3 Rooms Table

The Rooms table contains information regarding all available accommodations offered by Coravergel Inland Resort. It serves as the primary reference for room availability, pricing, occupancy limits, and reservation assignments.

Whenever a reservation is made, the system verifies room availability using this table before confirming the booking.

Table 4.6

Rooms Table Structure

Field Name	Data Type	Description
room_id	INT (Primary Key)	Room identification number
room_number	VARCHAR(20)	Assigned room number
room_type	VARCHAR(50)	Type of accommodation
capacity	INT	Maximum number of occupants
price	DECIMAL(10,2)	Room rate
status	VARCHAR(20)	Available, Reserved, Occupied,
Maintenance	description TEXT	Room description

The status field is automatically updated by the reservation module whenever bookings are created, modified, or completed.

4.7.4 Reservations Table

Field	Data type	Key	Description
reservation_id	INT, auto-increment	PK	Unique reservation identifier

customer_id	INT	FK	References Customers
user_id	INT	FK	References Users; the staff member who encoded the booking
check_in	DATE		Arrival date
check_out	DATE		Departure date
number_of_guests	SMALLINT		Guests in the party
status	ENUM		Pending, Confirmed, Checked-in, Checked-out, or Cancelled
total_amount	DECIMAL(10,2)		Computed total; recorded only, since online payment is out of scope
created_at	DATETIME		Time the reservation was created

4.7.5 Reservation Details Table

Field	Data type	Key	Description
detail_id	INT, auto-increment	PK	Unique line identifier
reservation_id	INT	FK	References Reservations
room_id	INT	FK	References Rooms
rate_applied	DECIMAL(10,2)		Nightly rate used for this booking
nights	SMALLINT		Computed from check-in and check-out
subtotal	DECIMAL(10,2)		rate_applied × nights

4.7.6 Activity Logs Table

Field	Data type	Key	Description
log_id	BIGINT, auto-increment	PK	Unique log identifier

user_id	INT	FK	References Users; the person who acted
action	VARCHAR(50)		Action performed, such as login, create, update, or cancel
record_affected	VARCHAR(100)		Table and record identifier acted upon
ip_address	VARCHAR(45)		Address the request came from
logged_at	DATETIME		Time of the action

4.8 User's Manual

This User's Manual provides a step-by-step guide for operating the Secure Web-Based Reservation Management System developed for Coravergel Inland Resort. It is intended for both administrators and customers to ensure proper and efficient use of the system. The manual explains the procedures from account registration, login, room reservation, reservation management, report generation, and secure logout. Following these procedures enables users to maximize the system's features while maintaining the security and integrity of reservation information.

4.8.1 Administrator User Guide

This Admin User Manual provides a complete guide to managing the CoraVergel Resort Admin Panel. Learn how to navigate key sections including the Dashboard, Bookings, Rooms, Notifications, and Settings. It covers essential tasks such as adding new rooms, editing room details, managing room availability, deleting room listings, and viewing the public website. This guide is designed to help administrators efficiently manage resort operations and room listings.

Step 1. Administrator Login

Procedure:

1. Open the CoraVergel Resort Reservation Management System using a web browser.
2. The login page will appear.
3. Enter the registered administrator username and password.
4. Click the Login button.

System Process:

- The system will validate the credentials you've entered.
- The password is checked against the password that is stored securely in the database.
- If the credentials are valid the Administrator Dashboard is displayed.
- If the credentials are incorrect, the system displays an error message.

Expected Output:

The administrator successfully logs in and gains access to the Dashboard.

Step 2. Dashboard

After successful login, the Dashboard is displayed. It provides an overview of the resort's reservation activities and system status.

The Dashboard may display:

- Total Reservations
- Available Rooms
- Occupied Rooms
- Pending Reservations
- Confirmed Reservations
- Total Customers
- Recent Activities

System Process:

- The system pulls the reservation, room and customer details from the database.
- The dashboard will show the updated information based on the records that are available.
- The navigation menu gives you access to the system admin functions.

Expected Output:

The administrator can monitor the resort's operations and navigate to the different management sections.

Step 3. Manage Reservations

Procedure:

1. Select Reservation Management from the navigation menu.
2. View the list of guest bookings.
3. Select a reservation to view the details of that reservation.
4. Review the guest information, selected room, check-in date, check-out date, and payment details, if available.
5. Use the available options to update the reservation status when necessary.

System Process:

- The system pulls the reservation records from the database.
- The reservation details are shown for your review.
- When an authorised update is submitted the system validates and saves the changes.
- The updated reservation status is reflected in the reservation list.

Expected Output:

The admin can view and manage guest bookings and can also monitor the status of those bookings.

Step 4. Manage Rooms

Procedure:

1. Select Room Management from the navigation menu.
2. View the list of registered resort rooms.

3. Click Add Room to create a new room listing, if available.
4. Enter the required room information, such as room name or number, room type, capacity, rate, description, and images.
5. Click Save or Add Room to submit the information.
6. To modify an existing room, select the available Edit option, update the details, and save the changes.

System Process:

- The system validates the required room information.
- The new room record is saved to the database.
- Updated room details are stored and reflected in the room list.
- The system displays an appropriate confirmation or error message.

Expected Output:

The administrator can add new rooms, update room information, and maintain accurate room listings.

Step 5. Manage Users

Procedure:

1. Select User Management from the navigation menu.
2. View the list of registered system users.
3. Click Add User to create a new staff or administrator account, if permitted.
4. Enter the required information, including full name, username, email address, contact number, user role, and password.
5. Click **Save** to create the account.
6. To modify an existing account, select the available Edit option and save the changes.

System Process:

- The system validates all required fields.
- The system checks for duplicate usernames or email addresses when applicable.
- The password is securely hashed before storage.
- The new or updated user information is saved to the database.

Expected Output:

The administrator can manage authorized user accounts and assign appropriate system roles.

Step 6. View Reports

Procedure:

1. Select Reports or the corresponding reporting option from the navigation menu.
2. Choose the available report category or date range.
3. Review the displayed reservation, room occupancy, customer, or revenue information.
4. Use the available export or print option, if supported.

System Process:

- The system retrieves relevant records from the database.
- The selected information is organized into a report.
- The system displays the report according to the selected filters.
- If supported, the report can be downloaded or printed.

Expected Output:

The administrator can review operational reports to monitor resort performance and support decision-making.

Step 7. View Activity Logs

Procedure:

1. Select Activity Logs from the navigation menu.
2. Review the recorded system activities.
3. Check the available details, such as the user, action performed, affected record, and date and time.
4. Apply available filters to locate specific activities, if supported.

System Process:

- The system retrieves recorded activities from the database.

- The activity records are displayed for authorized users.
- The administrator can review available logs to monitor system usage and account activity.

Expected Output:

The administrator can review recorded activities for monitoring, accountability, and security purposes.

Step 8. Manage Settings

Procedure:

1. Select Settings from the navigation menu.
2. Review the available system or account settings.
3. Update the required information or preferences.
4. Click Save or Update if available.

System Process:

- The system validates the submitted settings.
- Valid changes are saved to the database.
- An appropriate confirmation or error message is displayed.

Expected Output:

The administrator can manage the settings available to their account and maintain the required system preferences.

Step 9. View Website

Procedure:

1. Click View Website from the administrator interface, if available.
2. The public CoraVergel Resort website will open.
3. Review the displayed resort information, room listings, and reservation options.

System Process:

- The system opens the public-facing website.
- The website displays the available resort information and guest reservation features.

Expected Output:

The administrator can preview the public website and check how resort information is presented to guests.

Step 10. Administrator Logout

Procedure:

1. Locate the administrator profile or account menu.
2. Click Logout or Sign Out.
3. Confirm the action if the system requests confirmation.

System Process:

- The system ends the administrator's authenticated session.
- Protected administrative pages require authentication before they can be accessed again.
- The system redirects the administrator to the Login Page.

Expected Output:

The administrator is successfully logged out, helping protect the system from unauthorized access.

4.8.2 Guests User Guide

Step 1. Access the CoraVergel Resort Website

Procedure:

1. Open a web browser, such as Google Chrome.
2. Enter the CoraVergel Resort website address.
3. Wait for the homepage to load.
4. Browse the available resort information and reservation options.

System Process:

- The system loads the resort's public website.
- The homepage displays the available information and reservation features.
- The user can proceed to the booking section.

Expected Output:

The user successfully accesses the CoraVergel Resort website and is ready to begin the reservation process.

Step 2. Select Check-in and Check-out Dates

Procedure:

1. Locate the reservation or booking section on the homepage.
2. Select the preferred check-in date using the calendar.
3. Select the preferred check-out date.
4. Review the selected dates to ensure they are correct.

System Process:

- The system records the selected check-in and check-out dates.
- The selected dates establish the intended duration of the guest's stay.
- The system uses the selected dates when searching for available rooms.

Expected Output:

The user's preferred reservation dates are selected and ready for the next step.

Step 3. Select the Number of Guests and Rooms

Procedure:

1. Locate the guest and room selection options.
2. Select the number of rooms required.
3. Enter the number of adults.
4. Enter the number of children, if applicable.
5. Review the selected quantities.

System Process:

- The system records the selected number of rooms and guests.
- The selected information helps determine suitable accommodation options.
- The system prepares the booking search based on the user's selections.

Expected Output:

The number of rooms and guests is specified for the reservation.

Step 4. Search for Available Rooms

Procedure:

1. Review the selected dates and guest information.
2. Click the Book Now button.
3. Wait for the available room listings to appear.
4. Browse the available accommodations.

System Process:

- The system processes the submitted booking search.
- The system retrieves room information based on the selected reservation details.
- Available room listings and their corresponding information are displayed.

Expected Output:

The user can view the available accommodations that match the selected booking details.

Step 5. Choose a Preferred Room

Procedure:

1. Browse the available room listings.
2. Review the room photos, descriptions, rates, and other displayed details.
3. Compare the available options based on the user's needs and budget.
4. Select the preferred room.
5. Click the corresponding Book Now button.

System Process:

- The system displays the information associated with the selected room.
- The user's room selection is carried forward to the reservation process.
- The system proceeds to the guest information form.

Expected Output:

The user selects a preferred room and proceeds to provide the required booking information.

Step 6. Enter Guest Information and Upload Documents

Procedure:

1. Complete the required fields in the guest information form.
2. Review the entered personal and contact information.
3. Upload a valid identification document as required.
4. Upload the down payment receipt.
5. Check that the required information and uploaded files are correct.
6. Click Confirm Booking to submit the reservation request.

System Process:

- The system receives the submitted guest information.
- The system processes the uploaded identification document and down payment receipt.
- The system validates the required information and files according to its configured rules.
- The reservation request is submitted for processing.

Expected Output:

The user's reservation request is submitted with the required guest information and supporting documents.

Step 7. Review the Booking Submission and Wait for Confirmation

Procedure:

1. Wait for the system's response after submitting the booking.
2. Read the confirmation message displayed on the website.
3. Keep the registered email address and contact number accessible.
4. Monitor email or phone messages for reservation updates.

5. Wait for the resort's booking confirmation before treating the reservation as confirmed.

System Process:

- The system displays a message indicating that the booking request has been submitted.
- The reservation is processed according to the resort's booking procedures.
- The resort may contact the guest through the provided contact information regarding the reservation status.

Expected Output:

The user receives a submission acknowledgment and can monitor the provided contact channels for the resort's confirmation and further instructions.

REFERENCE

[1] K. Scarfone, M. Souppaya, and D. Dodson, “Secure Software Development Framework (SSDF) Version 1.1: Recommendations for Mitigating the Risk of Software Vulnerabilities,” National Institute of Standards and Technology, Gaithersburg, MD, USA, NIST Special Publication 800-218, Feb. 2022, doi: 10.6028/NIST.SP.800-218.

[2] C. Pascoe, S. Quinn, and K. Scarfone, “The NIST Cybersecurity Framework (CSF) 2.0,” National Institute of Standards and Technology, Gaithersburg, MD, USA, NIST Cybersecurity White Paper 29, Feb. 26, 2024, doi: 10.6028/NIST.CSWP.29.

[3] OWASP Foundation, “OWASP Top 10:2021,” 2021. [Online]. Available: https://owasp.org/Top10/

[4] OWASP Foundation, “OWASP Application Security Verification Standard (ASVS), Version 5.0.0,” May 2025. [Online]. Available: https://owasp.org/www-project-application-security-verification-standard/

[5] D. Temoshok, J. Fenton, Y.-Y. Choong, N. Lefkovitz, A. Regenscheid, R. Galluzzo, and J. Richer, “Digital Identity Guidelines: Authentication and Authenticator Management,” National Institute of Standards and Technology, Gaithersburg, MD, USA, NIST Special Publication 800-63B-4, Jul. 2025, doi: 10.6028/NIST.SP.800-63B-4.

[6] P. Fakfare, N. Manosuthi, J. S. Lee, M. Jin, H. Han, and J. J. Kim, "Data vulnerability and privacy risk among hotel guests who share personal data," *J. Vacation Mark.*, vol. 32, no. 2, pp. 390–407, Apr. 2026, doi: 10.1177/13567667241276213.

[7] B. L. Chua *et al.*, "Are you concerned about giving your private information to a hotel? Customer information privacy and its consequences," *Asia Pac. J. Tour. Res.*, Dec. 2025, early access, doi: 10.1080/10941665.2025.2608949.

[8] K. G. Yulius and K. Evelyn, "Exploring the use of PMS in a new hotel: A case study in Nio Garden Hotel Sumbawa," *Int. J. Travel, Hosp. Events*, vol. 4, no. 2, pp. 150–162, 2025, doi: 10.56743/ijothe.v4i2.529.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary of findings, conclusions, recommendations, impact of the study, and overall summary of the research entitled "Vulnerabilities of Manual Reservation Systems: A Secure Web-Based Reservation Management System for Coravergel Inland Resort." The study focused on identifying the operational and security issues associated with the resort's existing manual reservation process and developing a secure web-based reservation management system to address these challenges.

The developed system was designed to improve reservation processing, enhance data management, strengthen information security, and provide efficient reservation services through automation. By incorporating cybersecurity principles such as user authentication, role-based access control, password hashing, activity logging, and centralized database management, the proposed system minimizes the risks associated with manual reservation procedures while improving overall operational efficiency.

The following sections present the findings, conclusions, recommendations, and overall contributions of the study.

5.2 Summary of Findings

Based on the development and implementation of the proposed Secure Web-Based Reservation Management System, the researchers identified several significant findings that demonstrate the effectiveness of the developed application in addressing the vulnerabilities of the existing manual reservation system.[1]

The study found that the previous manual reservation process relied heavily on paper-based records and manual documentation, making reservation management time-consuming, error-prone, and vulnerable to security risks. Manual processing increased the possibility of duplicate bookings, misplaced reservation records, inaccurate customer information, delayed reservation confirmation, and inefficient report preparation.[2] Furthermore, the absence of centralized data management and proper access control exposed sensitive customer information to unauthorized access and limited the resort's ability to monitor reservation activities effectively.

To address these issues, the researchers successfully developed a Secure Web-Based Reservation Management System that automates reservation processing and centralizes the management of customer records, room availability, reservation transactions, and administrative reports. The system provides real-time monitoring of room availability, allowing reservation personnel to process bookings more efficiently while minimizing scheduling conflicts and reservation errors. The developed system also incorporates several security mechanisms to protect customer information and support secure system operations. These include user authentication, password hashing, role-based access control, session management, activity logging, and database backup procedures. These features improve accountability, protect confidential reservation data, and reduce the likelihood of unauthorized access or accidental data loss.

In addition, the system streamlines report generation by automatically organizing reservation records into a centralized database. Management can generate operational reports more quickly compared to the manual preparation of reservation logs, improving decision-making and administrative efficiency.

Overall, the findings indicate that the developed reservation management system effectively addresses the operational inefficiencies and security vulnerabilities identified in the existing manual reservation process while providing a more secure, reliable, and efficient platform for managing resort reservations.

5.3 Conclusion

Based on the objectives established in Chapter 1 and the results obtained throughout the development of the proposed system, the researchers conclude that the Secure Web-Based Reservation Management System successfully addresses the vulnerabilities associated with the manual reservation system of Coravergel Inland Resort.

The developed application significantly improves reservation management by automating reservation processing, centralizing customer information, monitoring room availability in real time, and generating accurate operational reports. Through automation, the system minimizes human errors commonly encountered in manual reservation processing, including duplicate bookings, inconsistent customer records, and delayed reservation confirmation.

From a cybersecurity perspective, the implementation of authentication mechanisms, role-based access control, password hashing, activity logging, and database backup procedures strengthens the protection of sensitive reservation information. These security features contribute to maintaining the confidentiality, integrity, and availability of customer data while promoting accountability among authorized personnel.[3]

The centralized database further enhances operational efficiency by allowing authorized users to retrieve reservation information quickly, improving communication among employees and reducing administrative workload. The system also supports better decision-making through organized reporting and real-time reservation monitoring.

Overall, the study demonstrates that replacing the traditional manual reservation process with a secure web-based information system can significantly improve operational performance, information security, and customer service within Coravergel Inland Resort. The developed system therefore serves as an effective technological solution that supports the resort's long-term operational and digital transformation objectives.

5.4 Recommendations

Although the developed reservation management system successfully addresses the identified operational and security challenges, the researchers recommend several enhancements that may further improve the functionality, security, and long-term sustainability of the application.[5]

5.4.1 System Improvements

To further enhance the performance and usability of the developed reservation management system, the following improvements are recommended:

- Integrate an online payment gateway to enable secure electronic payments.
- Implement SMS and email notification services for reservation confirmations, reminders, and cancellation notices.
- Develop a mobile-responsive or dedicated mobile application to improve accessibility for both administrators and customers.
- Add online customer self-service reservation capabilities, allowing guests to make reservations directly through the internet.
- Implement multi-factor authentication (MFA) for administrator accounts to strengthen user authentication.
- Incorporate automated backup scheduling and cloud-based storage to improve disaster recovery capabilities.
- Enhance reporting features by incorporating graphical analytics and occupancy trends to support management decision-making.

These enhancements will further improve operational efficiency while strengthening the overall security posture of the system.

5.4.2 Future Research Directions

Future researchers are encouraged to extend the capabilities of the proposed reservation management system by exploring additional technologies and security solutions.

Possible areas for future research include:

· Cloud-based reservation management systems.

· Artificial Intelligence (AI)-assisted reservation forecasting and occupancy analysis.

· Mobile reservation applications for Android and iOS platforms.

· Integration of QR Code technology for guest check-in and verification.

· Biometric authentication for administrator access.

· Automated cybersecurity monitoring and intrusion detection mechanisms.

· Integration with online travel booking platforms.

Future studies may also conduct comparative evaluations between manual reservation systems and fully automated reservation management platforms across different hospitality establishments.

5.4.3 Implementation Strategies

To ensure the successful adoption of the developed reservation management system, the following implementation strategies are recommended:

· Conduct comprehensive user training for administrators and reservation personnel before deployment.

· Establish standard operating procedures for system usage and reservation processing.

· Perform regular software maintenance and database optimization.

· Conduct periodic security assessments and vulnerability testing.

· Implement scheduled database backups and disaster recovery planning.

· Regularly update system passwords and user access privileges.

· Monitor system activity logs to identify suspicious activities and maintain accountability.

Proper implementation of these strategies will contribute to the long-term success and sustainability of the developed system.

5.5 System Improvements

To further enhance the performance and usability of the developed reservation management system, the following improvements are recommended:

· Integrate an online payment gateway to enable secure electronic payments.

· Implement SMS and email notification services for reservation confirmations, reminders, and cancellation notices.

· Develop a mobile-responsive or dedicated mobile application to improve accessibility for both administrators and customers.

· Add online customer self-service reservation capabilities, allowing guests to make reservations directly through the internet.

· Implement multi-factor authentication (MFA) for administrator accounts to strengthen user authentication.

· Incorporate automated backup scheduling and cloud-based storage to improve disaster recovery capabilities.

· Enhance reporting features by incorporating graphical analytics and occupancy trends to support management decision-making.

These enhancements will further improve operational efficiency while strengthening the overall security posture of the system.

5.6 Future Research Directions

Future researchers are encouraged to extend the capabilities of the proposed reservation management system by exploring additional technologies and security solutions.[4]

Possible areas for future research include:

· Cloud-based reservation management systems.

· Artificial Intelligence (AI)-assisted reservation forecasting and occupancy analysis.

· Mobile reservation applications for Android and iOS platforms.

· Integration of QR Code technology for guest check-in and verification.

· Biometric authentication for administrator access.

· Automated cybersecurity monitoring and intrusion detection mechanisms.

· Integration with online travel booking platforms.

Future studies may also conduct comparative evaluations between manual reservation systems and fully automated reservation management platforms across different hospitality establishments.

5.7 Implementation Strategies

To ensure the successful adoption of the developed reservation management system, the following implementation strategies are recommended:

· Conduct comprehensive user training for administrators and reservation personnel before deployment.

· Establish standard operating procedures for system usage and reservation processing.

· Perform regular software maintenance and database optimization.

· Conduct periodic security assessments and vulnerability testing.

· Implement scheduled database backups and disaster recovery planning.

· Regularly update system passwords and user access privileges.

· Monitor system activity logs to identify suspicious activities and maintain accountability.

Proper implementation of these strategies will contribute to the long-term success and sustainability of the developed system.

5.8 Impact of the Study

The successful implementation of the Secure Web-Based Reservation Management System is expected to provide significant benefits to Coravergel Inland Resort, its employees, customers, and future researchers.

For Coravergel Inland Resort, the system improves operational efficiency by automating reservation processes, reducing paperwork, minimizing reservation errors, and enhancing overall business productivity. Management gains better control over reservation operations through centralized data management and real-time reporting, supporting more informed decision-making.

For resort employees, the system simplifies daily reservation-related tasks by eliminating repetitive manual documentation and reducing administrative workload. Automated reservation processing allows staff members to focus more on customer service rather than record management, improving workplace productivity and efficiency.

For customers, the developed system enhances the reservation experience by providing faster reservation processing, more accurate booking information, and improved protection of personal data. Customers benefit from organized reservation handling and reduced scheduling conflicts, resulting in higher service quality and customer satisfaction.

From a cybersecurity perspective, the system contributes to protecting sensitive reservation information through secure authentication, role-based access control, password hashing, activity logging, and centralized database management. These mechanisms significantly reduce the risks associated with unauthorized access, data manipulation, and information loss.

Finally, the study provides future researchers with a practical reference for developing secure reservation management systems within the hospitality industry. The methodologies, security mechanisms, and system architecture presented in this study may serve as valuable foundations for future software development and cybersecurity research.

5.10 Summary

The study successfully identified the operational inefficiencies and security vulnerabilities associated with the manual reservation system of Coravergel Inland Resort and addressed these challenges through the development of a Secure Web-Based Reservation Management System.

By automating reservation processing and integrating modern cybersecurity mechanisms, the developed system enhances reservation accuracy, improves operational efficiency, strengthens information security, and supports better management decision-making. The incorporation of user authentication, password hashing, role-based access control, activity logging, and centralized database management demonstrates the importance of cybersecurity in protecting sensitive reservation information.

The researchers conclude that the developed system provides an effective and sustainable solution for improving reservation management while addressing the vulnerabilities inherent in manual reservation processes. With continuous maintenance and future enhancements, the system has the potential to further support the digital transformation initiatives of Coravergel Inland Resort and contribute to improved customer service, organizational efficiency, and information security.

5.11 REFERENCE

[1] R. Goecke, “The evolution of online booking systems,” in Handbook of e-Tourism. Cham, Switzerland: Springer, Sep. 2, 2022. [Online]. Available: https://doi.org/10.1007/978-3-030-48652-5_27.
[2] L. Florido-Benítez, “The cybersecurity applied by online travel agencies and hotels to protect users’ private data in smart cities,” Smart Cities, vol. 7, no. 1, pp. 475–495, Feb. 4, 2024, doi: 10.3390/smartcities7010019. [Online]. Available: https://doi.org/10.3390/smartcities7010019.
[3] OWASP Foundation, Application Security Verification Standard, ver. 5.0.0, May 2025. [Online]. Available: https://owasp.org/www-project-application-security-verification-standard/.
[4] A. A. Salameh, A. Al Mamun, N. Hayat, and M. H. Ali, “Modelling the significance of website quality and online reviews to predict the intention and usage of online hotel booking platforms,” Heliyon, vol. 8, no. 9, Art. no. e10735, Sep. 23, 2022, doi: 10.1016/j.heliyon.2022.e10735. [Online]. Available: https://doi.org/10.1016/j.heliyon.2022.e10735.
[5] D. Temoshok, Y.-Y. Choong, A. Regenscheid, R. Galluzzo, J. L. Fenton, J. Richer, and N. Lefkovitz, Digital Identity Guidelines: Authentication and Authenticator Management, NIST Special Publication 800-63B-4, Aug. 1, 2025, doi: 10.6028/NIST.SP.800-63B-4. [Online]. Available: https://doi.org/10.6028/NIST.SP.800-63B-4.
[6] PCI Security Standards Council, Payment Card Industry Data Security Standard: Requirements and Testing Procedures, ver. 4.0.1, Jun. 2024. [Online]. Available: https://www.pcisecuritystandards.org/document_library/.
[7] C. Pascoe, S. Quinn, and K. Scarfone, The NIST Cybersecurity Framework (CSF) 2.0, NIST Cybersecurity White Paper 29, Feb. 26, 2024, doi: 10.6028/NIST.CSWP.29. [Online]. Available: https://doi.org/10.6028/NIST.CSWP.29.

[8] A. H. Raghavendra, S. Bellary, P. K. Bala, and A. Mukherjee, “Exploring user adoption of online hotel booking platforms: A text mining approach,” Procedia Computer Science, vol. 234, pp. 106–113, 2024, doi: 10.1016/j.procs.2024.02.157. [Online]. Available: https://doi.org/10.1016/j.procs.2024.02.157